Neet Test Hub Premium Test Series 11
Answer Key

Physics

1 - A	2 - A	3 - D	4 - D	5 - B	6 - C	7 - D	8 - D	9 - C	10 - C
11 - C	12 - A	13 - C	14 - B	15 - A	16 - D	17 - A	18 - C	19 - D	20 - C
21 - B	22 - A	23 - D	24 - D	25 - B	26 - B	27 - B	28 - C	29 - D	30 - B
31 - C	32 - D	33 - B	34 - C	35 - C	36 - C	37 - D	38 - B	39 - A	40 - C
41 - D	42 - C	43 - B	44 - C	45 - B					

Chemistry

46 - B	47 - A	48 - A	49 - B	50 - A	51 - C	52 - D	53 - C	54 - C	55 - A
56 - B	57 - B	58 - C	59 - B	60 - D	61 - B	62 - D	63 - C	64 - A	65 - A
66 - A	67 - A	68 - B	69 - B	70 - D	71 - D	72 - B	73 - C	74 - B	75 - B
76 - A	77 - A	78 - B	79 - C	80 - A	81 - A	82 - A	83 - A	84 - D	85 - B
86 - B	87 - C	88 - D	89 - D	90 - C					

Biology - (Zoology)

91 - D	92 - C	93 - A	94 - C	95 - B	96 - D	97 - D	98 - D	99 - B	100 - D
101 - B	102 - A	103 - A	104 - B	105 - D	106 - D	107 - B	108 - B	109 - C	110 - C
111 - B	112 - A	113 - A	114 - B	115 - B	116 - D	117 - C	118 - A	119 - B	120 - B
121 - B	122 - C	123 - B	124 - A	125 - A	126 - B	127 - D	128 - A	129 - C	130 - D
131 - C	132 - B	133 - C	134 - C	135 - B					

Biology - (Botany)

136 - D	137 - B	138 - C	139 - A	140 - B	141 - A	142 - C	143 - A	144 - D	145 - B
146 - D	147 - D	148 - B	149 - C	150 - B	151 - D	152 - D	153 - B	154 - B	155 - A
156 - A	157 - A	158 - A	159 - A	160 - C	161 - C	162 - B	163 - B	164 - B	165 - B
166 - C	167 - D	168 - B	169 - C	170 - C	171 - A	172 - C	173 - A	174 - C	175 - D
176 - C	177 - A	178 - D	179 - D	180 - C					

Physics

1. Two wires A and B of the same material, having radii in the ratio $1:2$ carry currents in the ratio $4:1$. The ratio of drift speed of electrons in A and B is

A) $16:1$ B) $1:16$ C) $1:4$ D) $4:1$

Solution : (Correct Answer: A)

(a)

$4i = \text{neA}\ V_D$

$i = ne(4A)V_D'$

$4 = \frac{V_D}{4V_D'}$

$\frac{V_D}{V_D'} = 16:1$

2. The total momentum of electrons in a straight wire of copper of length $1\ metre$ carrying a current of $16\,A$ is

A) $91\times10^{-12}\,kg\,m/sec$ B) $91\times10^{-15}\,kg\,m/sec$
C) $91\times10^{-14}\,kg\,m/sec$ D) $91\times10^{-6}\,kg\,m/sec$

Solution : (Correct Answer: A)

$P = nMeV_d = \left[\frac{iL}{e}\right]\ Me$

3. A meta sample carrying a current along $x-$ axis with density J is subjected to a magnetic field B (along $z-$ axis). The electric field E developed along $y-$ axis is directly proportional to J as well as B. The constant of proportionality has $SI\ unit$

A) C/m^2 B) m^2s/C
C) m^2/C D) m^3/C

Solution : (Correct Answer: D)

$J = J = \frac{1}{A}$

$i = JA$

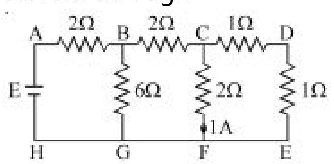
$\mathrm{ev_dB = eE}$

$\mathrm{ne\,Av_d = JA}$

$\mathrm{E = \frac{JB}{ne}}$

Constant of proporionality $= \frac{1}{\mathrm{ne}} = \frac{\mathrm{m^3}}{\mathrm{C}}$

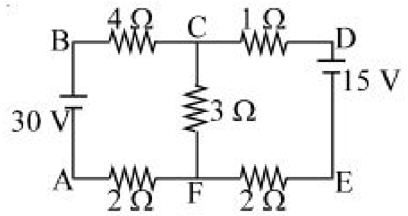
4. The figure shows a network of resistors and a battery. If $1\ A$ current flowsthrough the branch CF, then answer the following questions The current through



A) branch DE is $1\,A$
B) branch BC is $2\,A$
C) branch BG is $4\,A$
D) Both (A) and (B)

Solution : (Correct Answer: D)

5. The figure shows a network of five resistances and two batteries The current through the $15\ V$ battery is .... A



A) 0 B) 1 C) 3 D) none

Solution : (Correct Answer: B)

6. An electric kettle has two heating coils. When one of the coils is connected to an a.c. sotuce, the water in the kettle boils in 10 minutes. When the other coil is used the w'ater boils in 40 minutes. If both the coils are connected in parallel, the time taken by the same quantity of water to boil will be ...... min

A) 15 B) 25 C) 8 D) 4

Solution : (Correct Answer: C)

$Q = \frac{V^2}{R_1}\times t_1 = \frac{V^2}{R_2}\times t_2 = \frac{V^2}{R}\times t$

$\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R^2} \Rightarrow \frac{Q}{V^2t} = \frac{Q}{V^2t_1} + \frac{Q}{V^2t_2}$

$\Rightarrow \frac{1}{t} = \frac{1}{t_1} + \frac{1}{t_2} \Rightarrow t = \frac{t_1t_2}{t_1+t_2} = \frac{10\times40}{10+40}$

$= 8\ min$

7. A battery of $e.m.f.\ 10\ V$ and internal resistance $0.5\ ohm$ is connected across a variable resistance R. The value of R for which the power delivered in it is maximum is given by .... ohm

A) 2 B) 0.25 C) 1 D) 0.5

Solution : (Correct Answer: D)

For maximum power $r = R$

8. Thomson coefficient of a conductor is $10\ \mu V/K$. The two ends of it are kept at $50\,^oC$ and $60\,^oC$ respectively. Amount of heat absorbed by the conductor when a charge of $10\,C$ flows through it is

A) $1000\,J$ B) $100\,J$
C) $100\,mJ$ D) $1\,mJ$

Solution : (Correct Answer: D)

(d) By using $H = \sigma Q\theta$

$\Rightarrow H = (10\times10^{-6})\times10\times(60-50) = 10^{-3}J = 1\ mJ$

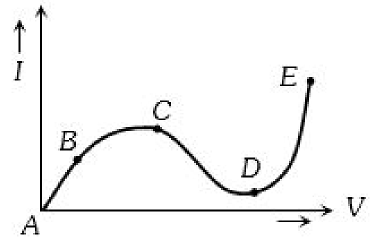
9. What is immaterial for an electric fuse wire

A) Its specific resistance B) Its radius
C) Its length D) Current flowing through it

Solution : (Correct Answer: C)

(c) Length is immaterial for an electric fuse wire.

10. From the graph between current I and voltage V shown below, identify the portion corresponding to negative resistance

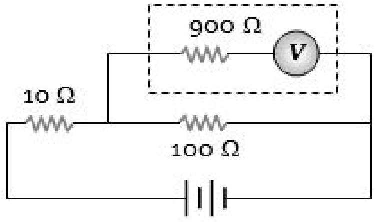


A) AB B) BC C) CD D) DE

Solution : (Correct Answer: C)

(c) For portion CD slope of the curve is negative i.e. resistance be negative.

11. The potential difference across the $100\ \Omega$ resistance in the following circuit is measured by a voltmeter of $900\ \Omega$ resistance. The percentage error made in reading the potential difference is



A) $\frac{10}{9}$ B) 0.1 C) 1 D) 10

Solution : (Correct Answer: C)

(c) Before connecting the voltmeter, potential difference across $100\,\Omega$ resistance

$V_i = \frac{100}{(100+10)} \times V = \frac{10}{11} V$

Finally after connecting voltmeter across $100\,\Omega$

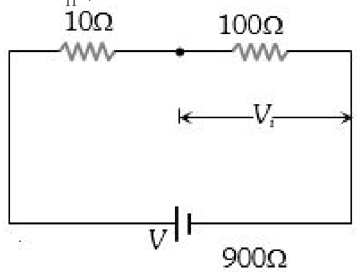
Equivalent resistance $\frac{100\times900}{(100+900)} = 90\,\Omega$

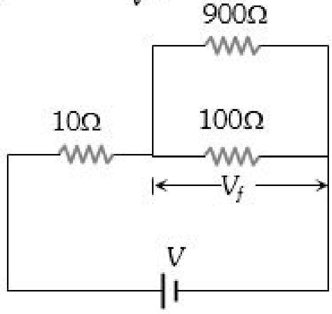
Final potential difference

$V_f = \frac{90}{(90+10)} \times V = \frac{9}{10} V$

% error = $\frac{V_i - V_f}{V_i} \times 100$

$= \frac{\frac{10}{11}V - \frac{9}{10}V}{\frac{10}{11}V} \times 100 = 1.0.$





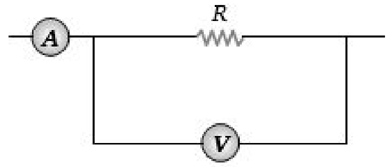
12. A Daniel cell is balanced on $125\ cm$ length of a potentiometer wire. Now the cell is short-circuited by a resistance $2\ ohm$ and the balance is obtained at $100\ cm$. The internal resistance of the Daniel cell is .... ohm

A) 0.5 B) 1.5 C) 1.25 D) 0.8

Solution : (Correct Answer: A)

(a) $r = \left(\frac{l_1 - l_2}{l_2}\right) R = \left(\frac{25}{100}\right) 2 = 0.5\,\Omega$

13. The ammeter A reads $2\ A$ and the voltmeter V reads $20\ V$. the value of resistance R is (Assuming finite resistance's of ammeter and voltmeter)



A) Exactly $10\,ohm$

B) Less than $10\,ohm$

C) More than $10\,ohm$

D) We cannot definitely say

Solution : (Correct Answer: C)

(c) $2R > 20 \Rightarrow R > 10\,\Omega.$

14. In an electrolyte 3.2×10^{18} bivalent positive ions drift to the right per second while 3.6×10^{18} monovalent negative ions drift to the left per second. Then the current is

A) $1.6\,amp$ to the left

B) $1.6\,amp$ to the right

C) $0.45\,amp$ to the right

D) $0.45\,amp$ to the left

Solution : (Correct Answer: B)

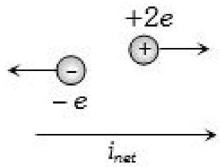
Net current $i_{net} = i_{(+)} + i_{(-)}$

$= \frac{n_{(+)}q_{(+)}}{t} + \frac{n_{(-)}q_{(-)}}{t}$

$= \frac{n_{(+)}}{t} \times 2e + \frac{n_{(-)}}{t} \times e$

$= 3.2 \times 10^{18} \times 2 \times 1.6 \times 10^{-19} + 3.6 \times 10^{18} \times 1.6 \times 10^{-19}$

$= 1.6\ A$ (towards right)



15. Current of $4.8\ A$ is flowing through a conductor. The number of electrons per second will be

A) 3×10^{19} B) 7.68×10^{21}

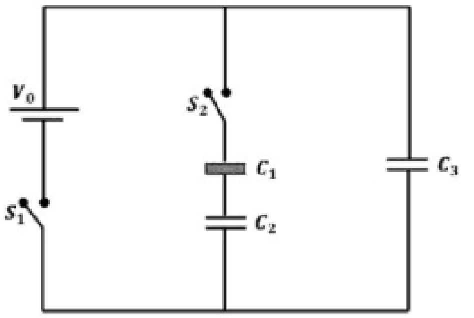
C) 7.68×10^{20} D) 3×10^{20}

Solution : (Correct Answer: A)

(a) Number of electrons flowing per second

$\frac{n}{t} = \frac{i}{e} = 4.8/1.6 \times 10^{-19} = 3 \times 10^{19}$

16. . Three identical capacitors C_1, C_2 and C_3 have a capacitance of $1.0\mu F$ each and they are uncharged initially. They are connected in a circuit as shown in the figure and C_1 is then filled completely with a dielectric material of relative permittivity ε_r. The cell electromotive force (emf) $V_0 = 8V$. First the switch S_1 is closed while the switch S_2 is kept open. When the capacitor C_3 is fully charged, S_1 is opened and S_2 is closed simultaneously. When all the capacitors reach equilibrium, the charge on C_3 is found to be $5\mu C$. The value of $\varepsilon_r =.....$



A) 1.40 B) 1.30 C) 1.20 D) 1.50

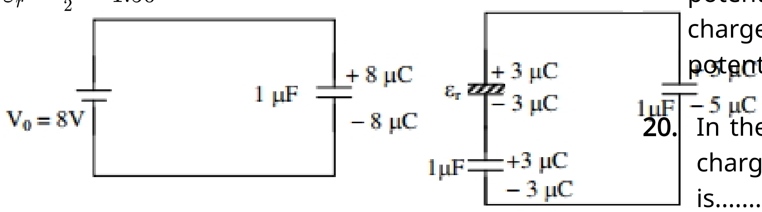
Solution : (Correct Answer: D)

Applying loop rule,

$\frac{5}{1} - \frac{3}{\varepsilon_r} - \frac{3}{1} = 0$

$2 - \frac{3}{\varepsilon_r} = 0$

$\varepsilon_r = \frac{3}{2} = 1.50$



17. Considering a group of positive charges, which of the following statements is correct?

A) Net potential of the system cannot be zero at a point but net electric field can be zero at that point.
B) Net potential of the system at a point can be zero but net electric field can't be zero at that point.
C) Both the net potential and the net field can be zero at a point.
D) Both the net potential and the net electric field cannot be zero at a point.

Solution : (Correct Answer: A)

18. Which of the following statements is true about the flow of electrons in an electric circuit?

A) Electrons always flow from lower to higher potential
B) Electrons always flow from higher to lower potential
C) Electrons flow from lower to higher potential, except through power sources
D) Electrons flow from higher to lower potential, except through power sources

Solution : (Correct Answer: C)

(c)

The free electrons experiences electrostatic force in the direction opposite to the direction of electric fired being is of negative charge. The electric field always directed from higher potential to lower potential. Therefore electrostatic force and negative charge or electrons always flows from lower to higher potential until the potentials become equal.

Hence, option (c) is correct.

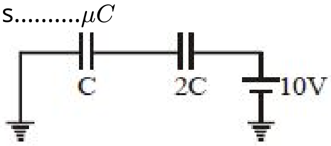
19. Assertion: Electron move away from a region of higher potential to a region of lower potential.
Reason: An electron has a negative charge.

A) If both Assertion and Reason are correct and the Reason is a correct explanation of the Assertion.
B) If both Assertion and Reason are correct but Reason is not a correct explanation of the Assertion.
C) If the Assertion is correct but Reason is incorrect.
D) If the Assertion is incorrect but the Reason is correct.

Solution : (Correct Answer: D)

Direction of electric field is from region of high potential to low potential & electron or any $-ve$ charged particle will move against the field or lower potential to higher potential.

20. In the circuit shown in the figure, $C = 6\ \mu F$. The charge stored in the capacitor of capacity C is..........μC



A) 0 B) 90 C) 40 D) 60

Solution : (Correct Answer: C)

$0 - \frac{Q}{C} + 10 - \frac{Q}{2C} = 0$

$Q = \frac{20C}{3} = \frac{20\times 6}{3} = 40\mu C$

21. If an electron enters into a space between the plates of a parallel plate capacitor at an angle α with the plates and leaves at an angle β to the plates, the ratio of its kinetic energy while entering the capacitor to that while leaving will be

A) $(\cos\alpha/\cos\beta)^2$
B) $(\cos\beta/\cos\alpha)^2$
C) $(\sin\alpha/\sin\beta)^2$
D) $(\sin\beta/\sin\alpha)^2$

Solution : (Correct Answer: B)

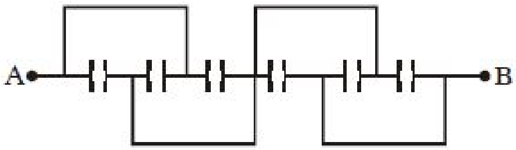
Let u be the velocity of electron while entering the field and v the velocity when it leaves the plates. Component of velocity parallel to plates will remain unchanged.

$u \cos\alpha = v \cos\beta \quad \therefore \frac{u}{v} = \frac{\cos\beta}{\cos\alpha}$

The desired ratio is.

$\frac{\frac{1}{2}mu^2}{\frac{1}{2}mv^2} = \left(\frac{u}{v}\right)^2 = \left(\frac{\cos\beta}{\cos\alpha}\right)^2$

22. All capacitors used in the diagram are identical and each is of capacitance C. Then the effective capacitance between the points A and B is



A) $1.5\,C$ B) $6\,C$ C) C D) $3\,C$

Solution : (Correct Answer: A)

First three capacitors are in parallel and last three are also in parallel. Both the groups are in series.

$\therefore C_{eq} = 3\mathrm{C}/2 = 1.5\mathrm{C}$

23. A charge Q is distributed over three concentric spherical shell of radii $a, b, c (a < b < c)$ such that their surface charge densities are equal to one another. The total potential at a point at distance r from their common centre, where $r < a$, would be

A) $\frac{Q}{12\pi \in_0} \frac{ab+bc+ca}{abc}$

B) $\frac{Q\,(a^2+b^2+c^2)}{4\pi \in_0 (a^3+b^3+c^3)}$

C) $\frac{Q}{4\pi \in_0 (a+b+c)}$

D) $\frac{Q\,(a+b+c)}{4\pi \in_0 (a^2+b^2+c^2)}$

Solution : (Correct Answer: D)

$\mathrm{Q_1 + q_2 + Q_3 = Q.........(1)}$

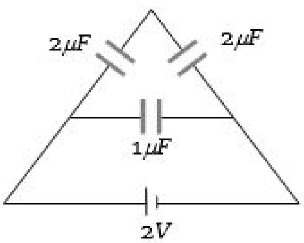
$\frac{\mathrm{Q_1}}{4\pi \mathrm{a}^2} = \frac{\mathrm{Q_2}}{4\pi \mathrm{b}^2} = \frac{\mathrm{Q_3}}{4\pi \mathrm{c}^2} = \mathrm{k.........(2)}$

Subs. Q_1, Q_2, Q_3 in (1)

$k = \frac{Q}{4\pi(a^2+b^2+c^2)}$

$v = \frac{kQ_1}{a} + \frac{kQ_2}{b} + \frac{kQ_3}{c}$

24. The charge on any one of the $2\,\mu\,F$ capacitors and $1\,\mu\,F$ capacitor will be given respectively (in $\mu\,C$) as

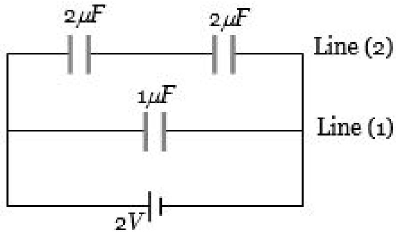


A) $1, 2$ B) $2, 1$ C) $1, 1$ D) $2, 2$

Solution : (Correct Answer: D)

(d) Potential difference across both the lines is same i.e. $2\,V$. Hence charge flowing in line 2

$Q = \left(\frac{2}{2}\right) \times 2 = 2\,\mu C$ So charge on each capacitor in line (2) is $2\,\mu C$



25. A hollow metallic sphere of radius R is given a charge Q. Then the potential at the centre is

A) Zero

B) $\frac{1}{4\pi\varepsilon_0} \cdot \frac{Q}{R}$

C) $\frac{1}{4\pi\varepsilon_0} \cdot \frac{2Q}{R}$

D) $\frac{1}{4\pi\varepsilon_0} \cdot \frac{Q}{2R}$

Solution : (Correct Answer: B)

(b) Potential V any where inside the hollow sphere, including the centre is $V = \frac{1}{4\pi\varepsilon_0} \frac{Q}{r}$.

26. Two metal spheres of radii R_1 and R_2 are charged to the same potential. The ratio of charges on the spheres is

A) $\sqrt{R_1} : \sqrt{R_2}$ B) $R_1 : R_2$

C) $R_1^2 : R_2^2$ D) $R_1^3 : R_2^3$

Solution : (Correct Answer: B)

(b) Spheres have same potential

i.e. $k\frac{Q_1}{R_1} = k\frac{Q_2}{R_2} ==> \frac{Q_1}{Q_2} = \frac{R_1}{R_2}$

27. The potential gradient at which the dielectric of a condenser just gets punctured is called

A) Dielectric constant

B) Dielectric strength

C) Dielectric resistance

D) Dielectric number

Solution : (Correct Answer: B)

we know The maximum electric field upto which an insulating (ori) non-conducting medium can mantain its insulating property is called the dielectric strength. potential Gradient $= \frac{dv}{dL}$ in direction of E.

28. Two electric charges $12\,\mu C$ and $-6\,\mu C$ are placed $20\,cm$ apart in air. There will be a point P on the line joining these charges and outside the region between them, at which the electric potential is zero. The distance of P from $-6\,\mu C$ charge is....... m

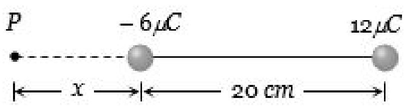
A) 0.10 B) 0.15 C) 0.20 D) 0.25

Solution : (Correct Answer: C)

(c) Point P will lie near the charge which is smaller in magnitude i.e. $-6\,\mu C$. Hence potential at P

$V = \frac{1}{4\pi\varepsilon_0} \frac{(-6\times10^{-6})}{x} + \frac{1}{4\pi\varepsilon_0} \frac{(12\times10^{-6})}{(0.2+x)} = 0$

$x = 0.2\,m$



29. A conductor with a positive charge

A) Is always at $+ve$ potential
B) Is always at zero potential
C) Is always at negative potential
D) May be at $+ve$, zero or $-ve$ potential

Solution : (Correct Answer: D)

(d) May be at positive, zero or negative potential, it is according to the way one defines the zero potential.

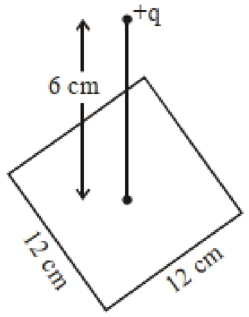
30. A proton is accelerated through $50,000\ V$. Its energy will increase by

A) $5000\,eV$ B) $8\times10^{-15}\,J$
C) $5000\,J$ D) $50,000\,J$

Solution : (Correct Answer: B)

(b) Kinetic energy $K = Q.V ==> K = (+e)(50000\,V)$
$= 50000\,eV = 50000\times1.6\times10^{-19}\,J = 8\times10^{-15}\,J$

31. A point charge of $+\ 12\ \mu C$ is at a distance $6\ cm$ vertically above the centre of a square of side 12 cm as shown in figure. The magnitude of the electric flux through the square will be .... $\times10^3$ Nm^2/C



A) 452 B) 381 C) 226 D) 113

Solution : (Correct Answer: C)

From symmetry $\phi = \frac{1}{6}\left(\frac{q}{\varepsilon_0}\right)$

$= \frac{12\times10^{-6}}{6\times8.85\times10^{-12}}$

$= 225.98\times10^3\ \frac{Nm^2}{s}$

$\simeq 226\times10^3\ \frac{Nm^2}{C}$

32. The electric field in a region is given $\vec{E} = \left(\frac{3}{5}E_0\hat{i} + \frac{4}{5}E_0\hat{j}\right)\frac{N}{C}$. The ratio of flux of reported field through the rectangular surface of area 0.2 m^2 (parallel to $y-z$ plane) to that of the surface of area $0.3\ m^2$ (parallel to $x-z$ plane) is $a:b$, where $a =$ ....
[Here $\hat{i},\hat{j}$ and $\hat{k}$ are unit vectors along x,y and $z-$ axes respectively]

A) 2 B) 3 C) 4 D) 1

Solution : (Correct Answer: D)

$\vec{E} = \left(\frac{3E_0}{5}\hat{i} + \frac{4E_0}{5}\hat{j}\right)\frac{N}{C}$

$A_1 = 0.2m^2$ [parallel to $y-z$ plane]

$= \vec{A}_1 = 0.2m^2\hat{i}$

$A_2 = 0.3m^2$ [parallel to $x-z$ plane]

$\vec{A}_2 = 0.3m^2\hat{j}$

Now $\phi_a = \left[\frac{3E_0}{5}\hat{i} + \frac{4E_0}{5}\hat{j}\right]\cdot[0.2\hat{i}] = \frac{3\times0.2}{5}E_0$

$\phi_b = \left[\frac{3E_0}{5}\hat{i} + \frac{4E_0}{5}\hat{j}\right]\cdot[0.3\hat{j}] = \frac{4\times0.3}{5}E_0$

Now $\frac{\phi_a}{\phi_b} = \frac{0.6}{1.2} = \frac{1}{2} = \frac{a}{b}$

$\Rightarrow a:b = 1:2$

$\Rightarrow a = 1$

33. A charged particle having some mass is resting in equilibrium at a height H above the centre of a uniformly charged non-conducting horizontal ring of radius R. The force of gravity acts downwards. The equilibrium of the particle will be stable R

A) for all values of H
B) only if $H > \frac{R}{\sqrt{2}}$
C) only if $H < \frac{R}{\sqrt{2}}$
D) only if $H = \frac{R}{\sqrt{2}}$

Solution : (Correct Answer: B)

$F = \frac{kQx(+q)}{(R^2+x^2)^{3/2}} - mg$

if $\frac{\mathrm{dF}}{\mathrm{dx}} < 0$

Then the particle is in stable equilibrium

$\Rightarrow \frac{(R^2+x^2)^{3/2} - \frac{3}{2}(R^2+x^2)^{1/2}(2x^2)}{(R^2+x^2)^3} < 0$

$\Rightarrow \mathrm{R}^2 + \mathrm{x}^2 - 3\mathrm{x}^2 < 0$

$\Rightarrow x > \frac{R}{\sqrt{2}}$ or $x < -\frac{R}{\sqrt{2}}$

$\therefore$ Only if $x > \frac{R}{\sqrt{2}}$

The equilibrium will be stable.

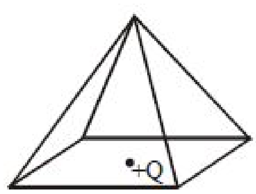
34. If a body is charged by rubbing it. Its weight

A) always decreases slightly
B) always increases slightly
C) may increase slightly or may decrease slightly
D) remains precisely the same

Solution : (Correct Answer: C)

Weight may increase or decrease depending uopn whether the body is charged negatively or positively.

35. A point charge $+Q$ is positioned at the centre of the base of a square pyramid as shown. The flux through one of the four identical upper faces of the pyramid is



A) $\frac{Q}{16\varepsilon_0}$ B) $\frac{Q}{4\varepsilon_0}$
C) $\frac{Q}{8\varepsilon_0}$ D) None of these

Solution : (Correct Answer: C)

Flux going in pyramid $= \frac{Q}{2\varepsilon_0}$

Which is divided equally among all 4 faces

$\therefore$ Flux through one face $= \frac{Q}{8\varepsilon_0}$

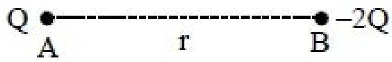
36. The point charges Q and $-2Q$ are placed at some distance apart. If the electric field at the location of Q is $\vec{E}$, then the electric field at the location of $-2Q$ will be :

A) $-\frac{\vec{E}}{2}$ B) $-\frac{3\vec{E}}{2}$
C) $+\frac{\vec{E}}{2}$ D) $-2\vec{E}$

Solution : (Correct Answer: C)

$E_A = E = \frac{K2Q}{r^2}$

$E_B = \frac{KQ}{r^2} = \frac{E}{2}$



37. Let $\rho(r) = \frac{Q}{\pi R^4} r$ be the charge density distribution for a solid sphere of radius R and total charge Q. For a point 'p' inside the sphere at distance r_1 from the centre of the sphere, the magnitude of electric field is

A) 0
B) $\frac{Q}{4\pi\varepsilon_0 {r_1}^2}$
C) $\frac{Q}{4\pi\varepsilon_0 R^4}$
D) $\frac{Q{r_1}^2}{3\pi\varepsilon_0 R^4}$

Solution : (Correct Answer: D)

Let us consider a spherical shell of thickness dx andradius x.

The volume of this spherical shell $= 4\pi r^2 dr$.

The charge enclosed within shell

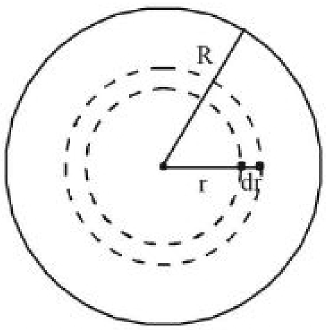
$= \frac{Qr}{\pi R^4}[4\pi r^2 dr]$

The charge enclosed in a sphere of radius r_1 is

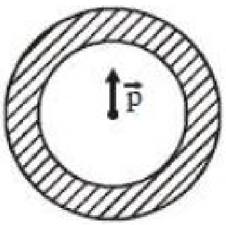
$= \frac{4Q}{R^4}\int_0^{r_1} r^3 dr = \frac{4Q}{R^4}\left[\frac{r^4}{4}\right]_0^r = \frac{Q}{R^4}r_1^4$

$\therefore$ The electric field at point p inside the sphere at a distance r_1 from the centre of the sphere is

$E = \frac{1}{4\pi\epsilon_0} - \frac{\left[\frac{Q}{R^4}r_1^4\right]}{r_1^2} = \frac{1}{4\pi\epsilon_0}\frac{Q}{R^4}r^2$



38. Shown in the figure is a shell made of a conductor. It has inner radius a and outer radius b, and carries charge Q. At its centre is a dipole $\vec{P}$ as shown. In this case



A) Surface charge density on the inner surface of the shell is zero everywhere
B) Electric field outside the shell is the same as that of a point charge at the centre of the shell
C) Surface charge density on the inner surface is uniform and equal to $\frac{(Q/2)}{4\pi a^2}$
D) Surface charge density on the outer surface depends on $\left|\vec{P}\right|$

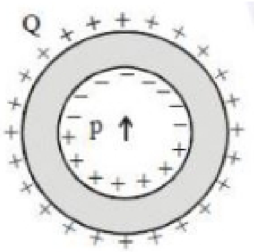
Solution : (Correct Answer: B)

Total charge of dipole $= 0$, so charge induced on outside surface $= 0$.

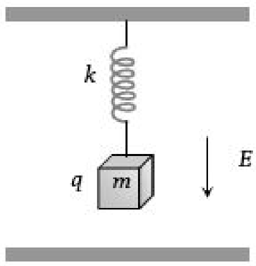
But due to non uniform electric field of dipole, the charge induced on inner surface is non zero and non uniform.

So, for any observer outside the shell, the resultant electric field is due to Q uniformly distributed on outer surface only and it is equal to.

$\boxed{E = \frac{KQ}{r^2}}$



39. Time period of a block suspended from the upper plate of a parallel plate capacitor by a spring of stiffness k is T. When block is uncharged. If a charge q is given to the block them, the new time period of oscillation will be



A) T B) $> T$
C) $< T$ D) $\geq T$

Solution : (Correct Answer: A)

(a) The forces that act on the block are qE and mg. Since qE and mg are constant forces, the only variable elastic force changes by kx. Where x is the elongation in the spring $\Rightarrow$ unbalanced (restoring) force $= F = -kx$

$\Rightarrow -m\omega^2 X = -KX \Rightarrow \omega = \sqrt{\frac{k}{M}} = T.$

40. The unit of electric permittivity is

A) $Volt/m^2$ B) $Joule/coulomb$
C) $Farad/m$ D) $Henry/m$

Solution : (Correct Answer: C)

(c) $C = \frac{\varepsilon_0 A}{d} \Rightarrow \varepsilon_0 = \frac{Cd}{A} \Rightarrow \varepsilon_0 \Rightarrow \frac{Farad \times m}{m^2} \to \frac{F}{m}$

41. One metallic sphere A is given positive charge whereas another identical metallic sphere B of exactly same mass as of A is given equal amount of negative charge. Then

A) Mass of A and mass of B still remain equal
B) Mass of A increases
C) Mass of B decreases
D) Mass of B increases

Solution : (Correct Answer: D)

(d) Negative charge means excess of electron which increases the mass of sphere B.

42. A charge q_1 exerts some force on a second charge q_2. If third charge q_3 is brought near, the force of q_1 exerted on q_2

A) Decreases
B) Increases
C) Remains unchanged
D) Increases if q_3 is of the same sign as q_1 and decreases if q_3 is of opposite sign

Solution : (Correct Answer: C)

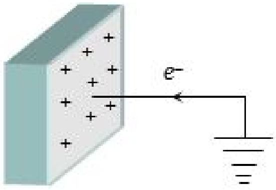
(c) The force will still remain $\frac{q_1 q_2}{4\pi\varepsilon_0 r^2}$

43. When a body is earth connected, electrons from the earth flow into the body. This means the body is.....

A) Unchanged
B) Charged positively
C) Charged negatively
D) An insulator

Solution : (Correct Answer: B)

(b) When a positively charged body connected to earth, electrons flows from earth to body and body becomes neutral.



44. An electron and a proton are in a uniform electric field, the ratio of their accelerations will be

A) Zero
B) Unity
C) The ratio of the masses of proton and electron
D) The ratio of the masses of electron and proton

Solution : (Correct Answer: C)

(c) $a = \frac{qE}{m} ==> \frac{a_e}{a_p} = \frac{m_p}{m_e}$

45. When a glass rod is rubbed with silk, it

A) Gains electrons from silk
B) Gives electrons to silk
C) Gains protons from silk
D) Gives protons to silk

Solution : (Correct Answer: B)

(b) On rubbing glass rod with silk, excess electron transferred from glass to silk. So glass rod becomes positive and silk becomes negative.

Chemistry

46. What is the two third life of a first order reaction having $\mathrm{k} = 5.48 \times 10^{-14} \mathrm{sec}^{-1}$?

A) 2.01×10^{11} s
B) 2.01×10^{13} s
C) 8.08×10^{13} s
D) 16.04×10^{11} s

Solution : (Correct Answer: B)

For two-third of a reaction ,

$[\mathrm{A}]_0 = \mathrm{a},\ [\mathrm{A}] = \mathrm{a} - \frac{2}{3}\mathrm{a} = \frac{\mathrm{a}}{3}$

$\mathrm{t}_{2/3} = \frac{2.303}{\mathrm{k}} \log \frac{[\mathrm{A}]_0}{[\mathrm{A}]}$

$= \frac{2.303}{\mathrm{k}} \log \frac{\mathrm{a}}{\frac{\mathrm{a}}{3}} = \frac{2.303}{\mathrm{k}} \log 3$

$\mathrm{t}_{2/3} = \frac{2.303 \times 0.4771}{5.48 \times 10^{-14}} = 2.01 \times 10^{13}$ s

47.

The half-life for decay of radioactive $^{14}\mathrm{C}$ is 5730 years. An archaeological artefact containing wood has only 80% of the $^{14}\mathrm{C}$ activity as found in living trees. The age of the artefact would be: [Given: log 1.25 = 0.0969]

A) 1845 years
B) 184.5 years
C) 1900 years
D) 190 years

Solution : (Correct Answer: A)

Radioactive decay follows first order kinetics.

Let $[\mathrm{A}]_0 = 100$

$\therefore\ [\mathrm{A}] = 100 \times 80\% = 80$

$\text{Decay constant (k)} = \frac{0.693}{\mathrm{t}_{1/2}} = \frac{0.693}{5730}$

$\mathrm{t} = \frac{2.303}{\mathrm{k}} \log \frac{[\mathrm{Ao}]}{[\mathrm{A}]}$

$= \frac{2.303}{\frac{0.693}{5730}} \log \frac{100}{80}$

$= \frac{2.303 \times 5730}{0.693} \times \log 1.25$

$= \frac{2.303 \times 5730}{0.693} \times 0.0969 = 1845 \text{ yrs}$

48. Fill in the blank :

$^{235}_{92}\mathrm{U} + {}^{1}_{0}\mathrm{n} \rightarrow ? + {}^{92}_{36}\mathrm{Kr} + 3{}^{1}_{0}\mathrm{n}$

A) $^{141}_{56}\mathrm{Ba}$
B) $^{139}_{56}\mathrm{Ba}$
C) $^{139}_{54}\mathrm{Ba}$
D) $^{141}_{54}\mathrm{B}$

Solution : (Correct Answer: A)

$92 + 0 = \mathrm{Z} + 36 + 0 \Rightarrow \mathrm{Z} = 56$ (Sum total atomic numbers)

$235 + 1 = \mathrm{A} + 92 + 3$ (Sum total mass numbers)

$\therefore\ \mathrm{A} = 141$

Missing nuclide is $^{141}_{56}\mathrm{Ba}$

49. For the reaction,

$$2N_2O_5 \rightarrow 4\ NO_2 + O_2$$

the values of rate constant and rate of reaction are respectively $3\ \times 10^{-5} s^{-1}$ and $2.4 \times 10^{-5} mol\ L^{-1} s^{-1}$. The concentration of N_2O_5 (in mol L^{-1}) will be:

A) 0.4 B) 0.8 C) 1.2 D) 0.6

Solution : (Correct Answer: B)

Since, unit of rate constant is $3.0 \times 10^{-5} s^{-1}$, means order of reaction is one.

Thus, our rate law will be: $r = k\,[N_2O_5]$

$k = 3 \times 10^{-5}$

$\therefore [N_2O_5] = \frac{r}{k} = \frac{2.4\times10^{-5}}{3\times10^{-5}} = 0.8\ \ mol\ \ L^{-1}$

50. Which is incorrect statement in the following?

A) α -rays have more penetrating power than β -rays

B) α -rays have less penetrating power than γ -rays

C) β -rays have less penetrating power than γ -rays

D) β -rays have more penetrating power than α -rays

Solution : (Correct Answer: A)

The order of penetrating power is : $\alpha < \beta < \gamma$ -rays. It is due to lower mass and high speed.

51. The mechanism of the reaction, $2NO\,(g) + 2H_2\,(g) \rightarrow N_2\,(g) + 2H_2O\,(g)$, is

Step 1: $2NO\,(g) + H_2\,(g) \xrightarrow{\text{slow}} N_2 + H_2O_2$

Step 2: $H_2O_2 + H_2 \xrightarrow{\text{fast}} 2H_2O$

Select the correct statement

A) rate $= k[NO]^2[H_2]^2$

B) rate $= k\,[H_2O_2]\,[H_2]$

C) on doubling the concentration of H_2 , keeping the concentration of NO constant, the rate will become double

D) if the initial concentration of H_2 and NO is C_0 and after time 't' the concentration of N_2 is x, then rate $= k(C_0 - 2x)^x$

Solution : (Correct Answer: C)

As slowest step is the rate determining.

Rate $= k[NO]^2\,[H_2]$

So on doubling the concentration of H_2 without changing the concentration of NO, the rate will become double.

52. The temperature coefficient of a reaction is 2. When the temperature is increases from $30\,°C$ to $90°C$, the rate of reaction is increased by

A) 150 times B) 410 times C) 72 times D) 64 times

Solution : (Correct Answer: D)

Temperature coefficient 2 means rate of reaction doubles at every $10°C$ rise in temperature.

Thus, $\frac{k_{90°C}}{k_{30°C}} = 2^6 = 64$

53. When a bio-chemical reaction is carried out in laboratory, outside the human body in absence of enzyme, then rate of reaction obtained is 10^{-6} times, the activation energy of reaction in the presence of enzyme is

A) -6 RT B) +6 RT

C) -6 (2.303) RT D) +6 (2.303) RT

Solution : (Correct Answer: C)

We know that,

Rate of reaction(K) $= Ae^{-E/RT}$ ...(i)

And, $10^6.K = Ae^{-E_c/RT}$ ...(ii)

By dividing equation (ii) by (i)

$\frac{10^6.K}{K} = \frac{Ae^{-E_c/RT}}{Ae^{-E/RT}}$

$10^6 = e^{(E-E_c)/RT}$

$6.\ln 10 = (E - Ec)\,/RT$

$\frac{E-E_c}{RT} = 2.303 \times 6$

$E - E_c = 6 \times 2.303RT$

or $\Delta E_a = E_c - E = -6 \times 2.303RT$

54. Which of the given radiations is having high penetrating power and not affected by electrical and magnetic field?

A) Alpha rays B) Beta rays

C) Gamma rays D) Neutrons

Solution : (Correct Answer: C)

γ Gamma-rays is form of energy from radioactive decay, have maximum penetrating power.

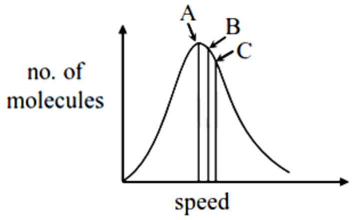
55. Calculate the average life (in minutes), if the half-life of a radionuclide is 69.3 minutes.

A) 100 B) $1/100$ C) 69.3×14.4 D) 0.693×69.3

Solution : (Correct Answer: A)

Average life $(\tau) = 1.44t_{1/2} = 1.44 \times 69.3 = 99.7 \approx 100$ minutes.

56. Identify the correct labels of A, B and C in the following graph from the option given below



Root mean square speed (V_{rms}) ; most probable speed (V_{mp}) ; Average speed (V_{av})

A) $A–V_{rms}; B–V_{mp}; C - V_{av}$

B) $A - V_{mp}; B - V_{av}; C - V_{rms}$

C) $A - V_{mp}; B - V_{rms}; C - V_{av}$

D) $A - V_{av}; B - V_{rms}; C - V_{mp}$

Solution : (Correct Answer: B)

As on increasing temperature velocities increase but fraction of molecules having these velocities decrease. As V_{rms} is highest velocity but possessed by least number of molecules and V_{mps} is least velocity but possessed by maximum molecules, hence $A - V_{mp}; B - V_{av}; C - V_{rms}$

57. The half life period for catalytic decomposition of XY_3 at 100 mm is found to be 8 hrs and at 200 mm it is 4 hrs. The order of reaction is

A) 3 B) 2 C) 4 D) None of these

Solution : (Correct Answer: B)

The value of order of reaction using half life and pressure values can be find out by using this relation given below

$\frac{(t_{\frac{1}{2}})_1}{(t_{\frac{1}{2}})_2} = \left(\frac{P_2}{P_1}\right)^{n-1}$

or $\frac{8}{4} = \left(\frac{200}{100}\right)^{n-1}$

$(2)^1 = (2)^{n-1}$

or $1 = n - 1$

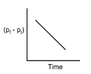
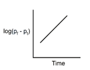
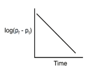
So, $n = 2$.

Hence, it is second order reaction.

58. Benzene diazonium chloride in aqueous solution decomposes as :

$C_6H_5 - N = N^+Cl^-_{(aq)} + H_2O_{(aq)} \longrightarrow C_6H_5OH_{(aq)} + N_{2(g)} + HCl_{(aq)}$

The reaction follows first order kinetics. If p_t is the pressure of N_2 at constant volume and temperature corresponding to different intervals of time t and p_f that after completion of the reaction, then which of the following graphs conforms to the kinetics of the reaction ?

A)  B)  C)  D)

Solution : (Correct Answer: C)

$p_f \propto$ Initial moles of $C_6H_5N_2^+Cl^-$ (a)

$p_t \propto$ Moles of N_2 evolved at time t

$\propto$ Moles of $C_6H_5\,N_2^+Cl^-$ reacted

$(p_f - p_t) \propto$ mole of $C_6H_5N_2^+Cl^-$ left at time t (a - x)

msub $= 2.303 \log \frac{a}{a - x}$

msub $t = 2.303 \log \frac{p_f}{(p_f - p_t)}$

$\frac{kt}{2.303} = \log P_f - \log(P_f - P_t)$

$\log(p_f - p_t) = \log p_f - \frac{k}{2.303} t$

Interception y axis = log p_f

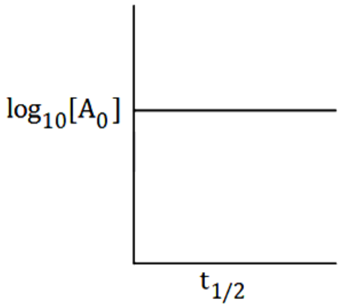
$\text{Slope} = -\frac{k}{2.303}$

59. Plotting $\log_{10} t_{1/2}$ against $\log_{10}[A_0]$ (where A_0 is the initial concentration of a reactant) for a first order reaction the slope will be

A) -2 B) Zero C) $+1$ D) -1

Solution : (Correct Answer: B)

The $t_{1/2}$ of a first order reaction is independent of the initial concentration of the reactants.



60. The rate law for the chemical reaction

$2NO_2Cl \rightarrow 2NO_2 + Cl_2 = k[NO_2Cl]$ is rate The rate determining step is

A) $2NO_2Cl \rightarrow 2NO_2 + 2Cl$ B) $NO_2 + Cl_2 \rightarrow NO_2Cl + Cl$

C) $NO_2Cl + Cl \rightarrow NO_2 + Cl_2$ D) $NO_2Cl \rightarrow NO_2 + Cl$

Solution : (Correct Answer: D)

Rate $= k[NO_2Cl]$

Hence ,rate determtextning step is

$NO_2Cl \rightarrow NO_2 + Cl$

61. For the cell reaction,

$\underset{(C_1)}{Cu^{2+}(aq)} + Zn(s) \rightarrow \underset{(C_2)}{Zn^{2+}(aq)} + Cu(s)$,

the change in free energy (ΔG) at a given temperature is a function of

A) $\ln C_1$ B) $\ln(C_2/C_1)$ C) $\ln(C_1 + C_2)$ D) $\ln C_2$

Solution : (Correct Answer: B)

$\Delta G = \Delta G^o + RT \ln Q$

$\Delta G = \Delta G^o + RT \ln \frac{[Zn^{+2}]}{[Cu^{+2}]}$

$\Delta G = \Delta G^o + RT \ln \frac{C_2}{C_1}$

So, ΔG is function of $\ln(C_2/C_1)$.

62. Three Faraday of electricity is passed through aqueous solutions of $AgNO_3, NiSO_4$ and $CoCl_3$ kept in three vessels using inert electrodes. The ratio in mol in which the metals Ag, Ni and Co will be deposited is

A) 3:2:1 B) 1:2:3 C) 2:3:6 D) 6:3:2

Solution : (Correct Answer: D)

(i) $Ag^+(aq) + \underset{\substack{1\text{ mol} = 1\text{ F} \\ 3\text{ F}}}{e^-} \rightarrow \underset{\substack{1\text{ mol} \\ 3\text{ mol}}}{Ag\ (s)}$

(ii) $Ni^{2+}(aq) + \underset{\substack{2\text{ mol} = 2\text{ F} \\ 3\text{ F}}}{2e^-} \rightarrow \underset{\substack{1\text{ mol} \\ 3/2\text{ mol}}}{Ni\ (s)}$

(iii) $Co^+(aq) + \underset{3\text{ mol} = 3\text{ F}}{3e^-} \rightarrow \underset{1\text{ mol}}{Co\ (s)}$

The required ratio of moles of Ag, Ni and Co is 3 mol Ag: 3/2 mol Ni : 1 mol Co or 6 mol Ag : 3 mol Ni : 2 mol Co.

63. The equivalent conductivity of 0.1 M weak acid is 100 times less than that at infinite dilution. The degree of dissociation of weak electrolyte at 0.1 M is

A) 100 B) 10 C) 0.01 D) 0.001

Solution : (Correct Answer: C)

$\Lambda_v = \frac{\Lambda^\circ}{100}$

So $\alpha = \frac{\Lambda_v}{\Lambda^\circ} = \frac{\Lambda^\circ}{100\Lambda^\circ} = 0.01$

64. An electrolytic cell is constructed for preparing hydrogen. For an average current of 2 ampere in the circuit, the time required to produce 450 mL of hydrogen at NTP is approximately

A) 1/2 hour B) 1 hour C) 2 hours D) 5 hours

Solution : (Correct Answer: A)

$22400\ \mathrm{mL}\ H_2$ at STP $= 1\ \mathrm{mol}\ H_2$

So $450\ \mathrm{ml}\ H_2$ at STP

$= \frac{450\ \mathrm{mL}}{22400\ \mathrm{mL}} \times 1\ \mathrm{mol}\ H_2$

$= 0.02\ \mathrm{mol}$

Electrode reaction

$\underset{2F}{2H^+ + 2e^-} \rightarrow \underset{1mol}{H_2}$

The liberation of $1\ \mathrm{mol}\ H_2$ requires $= 2 \times 96500\ \mathrm{C}$

So the liberation of $0.02\ \mathrm{mol}\ H_2$ requires

$= 2 \times 96500 \times 0.02\ \mathrm{C}$

$= 3860$ C

Time required to produce $0.02\ \mathrm{mol}\ H_2$, $t = \frac{Q}{i}$

$= \frac{3860}{2A} = 1930\ \mathrm{s} \simeq 0.503\ \mathrm{hour}$

$\approx 0.5\ \mathrm{hour}$

65. The conductivity of strong electrolyte

A) Increases on dilution slightly
B) Decreases on dilution
C) Does not change with dilution
D) Depends upon density of electrolyte itself

Solution : (Correct Answer: A)

Strong electrolytesThese well-behaved systems include many simple salts such as NaCl, as well as all strong acids. The Λ vs. √c plots closely follow the linear relationΛ = Λ° – b √c

66. Calculate the volume of O_2 liberated at STP by passing 5A current for 193 sec through acidified water:

A) 56 mL B) 112 mL
C) 158 mL D) 965 mL

Solution : (Correct Answer: A)

As per the first law of electrolysis, The mass deposited or volume liberated at the electrode is directly proportional to the charge passing between the electrode.

$m = ZQ$

Where $Z = \frac{E}{F}$ (electrovalent equivalent)

$Q = i \times t.$

mass of $O_2 = \frac{E \times i \times t}{F}$ (Where $F = 96500$ C)

$= \frac{8 \times 5 \times 193}{96500}\ \mathrm{g}$

$= 0.08\ \mathrm{g}$

$\because$ At STP, volume of $32\ \mathrm{g}\ O_2 = 22400\ \mathrm{mL}$

$\therefore$ The volume of $0.08\ \mathrm{g}\ O_2 = \frac{22400 \times 0.08}{32}$

$= 56\ \mathrm{ml}$

67. Given below are the half - cell reactions :

$$Mn^{2+} + 2e^- \rightarrow Mn ; \quad E^\circ = -1.18\ V$$
$$2\left(Mn^{3+} + e^- \rightarrow Mn^{2+}\right) ; \quad E^\circ = +1.51\ V$$

The E° for $3Mn^{2+} \rightarrow Mn + 2Mn^{3+}$ will be :

A) -2.69 V ; the reaction will not occur
B) -2.69 V ; the reaction will occur
C) -0.33 V ; the reaction will not occur
D) -0.33 V ; the reaction will occur

Solution : (Correct Answer: A)

$\Delta G^\circ = -nFE^\circ$

$Mn^{+2} + 2e^- \rightarrow Mn$ .... (1)

$\Delta G = -2 \times F \times (-1.18) = +2.36\ F$

$2Mn^{+3} + 2e^- \rightarrow 2Mn^{+2}$ .... (2)

$\Delta G = -2 \times F \times (-1.51) = +3.02\ F$

(1) + (2)

$3Mn^{+2} \rightarrow Mn + 2Mn^{+3}$

$\Delta G = +2.36F + (3.02F) = 5.38\ F$

$\Delta G = -2 \times F \times E$

$E = \frac{5.38F}{-2 \times F} = -2.69\ V$

We know that when $E_{cell} < 0$, cell is non-spontaneous.

68. Mark out the correct statement among the following.

A) Aqueous $AgNO_3$ solution can be stored in a copper bowl.
B) Aqueous $CuSO_4$ solution can be stored in a silver bowl.
C) Cu , Ag can release hydrogen gas from dil HCl .
D) H_2 cannot reduce Cu^{2+} and Ag^+ in the form of metals.

Solution : (Correct Answer: B)

Aq $CuSO_4$ Solution can be stored in a silver bowl because Ag is less reactive than Cu

69. The equivalent conductance of 1M benzoic acid is $12.8\ \mathrm{ohm^{-1}cm^2}$ and if the conductance of benzoate ion and H^+ ion at infinite dilution are 42 and $288.42\ \mathrm{ohm^{-1}cm^2}$ respectively. Then its degree of dissociation is

A) 0.039% B) 3.9% C) 0.35% D) 39%

Solution : (Correct Answer: B)

$\Lambda^{o}_{m(C_6H_5COOH)} = \Lambda^{o}_{(C_6H_5COO^-)} + \Lambda^{o}_{(H^+)}$
$= 42 + 288.42 = 330.42$
$\alpha = \frac{\Lambda^c_m}{\Lambda^o_m} = \frac{12.8}{330.42} = 3.9\%$

70. A silver cup is plated with silver by passing 965 C of electricity. The amount of Ag deposited is

A) 107.89 g B) 9.89 g C) 1.0002 g D) 1.08 g

Solution : (Correct Answer: D)

$Ag^{+} + e^{-} \longrightarrow Ag$

∵ 96500 C are required to deposite Ag = 108 g

∴ 965 C are required to deposite Ag

$= \frac{108}{96500} \times 965 = 1.08\ g$

71. The EMF of a cell corresponding to the reaction :

$Zn(s) + 2H^{+}(aq) \rightarrow Zn^{2+}(0.1\ M) + H_2(g)\ (1\ atm.)$ is 0.28 volt at 25°C.

The pH of the solution at the hydrogen electrode.

$E^{o}_{Zn^{2+}/Zn} = -0.76$ volt; $E^{o}_{H^+/H_2} = 0$

A) 2.30 B) 7.8 C) 9.2 D) 8.30

Solution : (Correct Answer: D)

The cell may be represented as

$Zn(s)\ |Zp^2\ (aq)|\ |H\ (aq)|\ H_2(g)1atmPt(s)$

$Zn\ (s) \rightarrow Zn^{2+}\ (aq) + 2\bar{e}$ (Anode)
$2H^{+}\ (aq) + 2\bar{e} \rightarrow H_2(g)\ 1atm$ (Cathode)

$Zn\ (s) + 2H^{+}\ (aq) \xrightarrow{2\bar{e}} Zn^{2+}\ (aq) + H_2(g)\ 1atm$ (CellReaction)

$E_{cell} = 0.28 = E^{o}_{cell} - \frac{0.059}{2}\log\frac{[Zn^{2+}_{aq}]}{[H^+]^2}$

$0.28 = 0 - (-0.76) - \frac{0.059}{2}\ \log\ \frac{0.1}{[H^+]^2}$

$0.28 - 0.76 = -0.48 = \frac{-0.059}{2}\ \log\ \frac{0.1}{[H^+]^2}$

$\frac{0.48\times2}{0.059} = 16.27 = \ \log\ \frac{0.1}{[H^+]^2}$

$\frac{0.1}{[H^+]^2} = 1.862 \times 10^{+16}$

$[H^+]^2 = \frac{0.1}{1.862\times10^{+16}}$

$= 0.0537 \times 10^{-16/2}$

$[H^+] = 5.37 \times 10^{-9}$

pH = - log (5.37 x 10 -9)

= 9 - log 5,37

= 9 - 0.73 = 8.27

≈ 8.3

72. 200 ml of water is added to 500 ml of 0.2 M solution. What is the molarity of this diluted solution?

A) 0.7093 M B) 0.1428 M C) 0.4005 M D) 0.2897 M

Solution : (Correct Answer: B)

No. of millimoles $= 500 \times 0.2 = 100$

Total volume in mililitre = 500 + 200 = 700

Thus, molarity of diluted Solution: $= \frac{100}{700} = 0.1428\ M$

73. The reactions that occurs at the cathode of a common dry cell is

A) $2ZnO_2 + Mn^{2+} + 2e^- \rightarrow MnZn_2O_4$
B) $Mn \rightarrow Mn^{2+} + 2e^-$
C) $2MnO_2 + Zn^{2+} + 2e^- \rightarrow ZnMn_2O_4$
D) $Zn \rightarrow Zn^{2+} + 2e^-$

Solution : (Correct Answer: C)

In common dry cell.

Anode: $Zn \rightarrow Zn^{++} + 2e^-$

Cathode: $2MnO_2 + Zn^{++} + 2e^- \rightarrow ZnMn_2O_4$.

74. Given that E ° values of Ag^+/Ag, K^+/K, Mg^{2+}/Mg and Cr^{3+}/Cr are $0.80V$, $-2.93V$, $-2.37V$ and $-0.74V$ respectively. Therefore the order for the reducing power of the metal is -

A) $Ag > Cr > Mg > K$
B) $Ag < Cr < Mg < K$
C) $Ag > Cr > K > Mg$
D) $Cr > Ag > Mg > K$

Solution : (Correct Answer: B)

More the negative E° value, larger the reducing power of the metal.

75. The material used to make anode and cathode of the commonly employed dry cells are

A) aluminium and lead
B) zinc and carbon
C) an alloy of silver, copper and manganese
D) brass and a mixture of ammonium chloride and manganese

Solution : (Correct Answer: B)

Zinc → anode

Carbon → cathode

76. The vapour pressure of pure benzene at 25 °C is 640 mm Hg and that of the solute A in benzene is 630 mm of Hg. The molality of solution is

A) 0.2 m B) 0.4 m C) 0.5 m D) 0.1 m

Solution : (Correct Answer: A)

$P = P^{o}X_A$
$630 = 640X_A$
$X_A = \frac{630}{640} = 0.984$
$X_B = 0.016$
$m = \frac{X_B\times1000}{X_AM_A} = \frac{0.016\times1000}{0.984\times78}$
$= 0.2\ m$

77. 0.01 M solution of KCl and $CaCl_2$ are separately prepared in water. The freezing point of KCl is found to be -2^oC . What is the freezing point of $CaCl_2$ aq. solution if it is completely ionized?

A) -3^oC B) $+3^oC$ C) -2^oC D) -4^oC

Solution : (Correct Answer: A)

$\mathrm{i\ for\ KCl = 2,\ i\ for\ for\ CaCl_2 = 3}$

$\Delta T_f \propto i$

$\frac{\Delta T_f(KCl)}{\Delta T_f(CaCl_2)} = \frac{2}{3}$

$\Delta T_f\,(CaCl_2) = \frac{3}{2} \times 2 = 3^oC$

Freezing point of $CaCl_2 = -3^oC$

78. Which substance would give a solution with a boiling point below that of pure water rather than above ?

A) Sodium chloride (solid)
B) Ethyl alcohol (liquid, b.p. 61°C)
C) Sulphuric acid (liquid, b.p. > 300°C)
D) Sucrose sugar (solid)

Solution : (Correct Answer: B)

When two liquids are mixed, the boiling point of the mixture lies somewhere between the boiling points of the two pure substances. Since ethyl alcohol has a lower boiling point than water, a mixture of the two would also have a lower boiling point.

79. Which one of the following electrolytes has the same value of van't Hoff's factor (i) as that of the $Al_2(SO_4)_3$ (if all are 100% ionised)?

A) K_2SO_4 B) $K_3[Fe(CN)_6]$
C) $K_4[Fe(CN)_6]$ D) $Al(NO_3)_3$

Solution : (Correct Answer: C)

van't Hoff factor of $Al_2(SO_4)_3 \rightarrow 2Al^{+3} + 3SO_4^{-2}$

So $n = 5$

$K_4[Fe(CN)_6] \rightarrow 4K^+ + [Fe(CN)_6]^{-4}$

So $n = 5$

As $\alpha = 100\% = 1$

$i = n = 5$

In K_2SO_4 i = 2, In $K_3[Fe(CN)_6]$ i = 4, In $Al(NO_3)_3$ i = 4

80. At 35°C, the vapour pressure of CS_2 is 512 mm Hg and that of acetone is 344 mm Hg. A solution of CS_2 in acetone has a total vapour pressure of 600 mm Hg. The false statement amongst the following is

A) A mixture of 100 mL CS_2 and 100 mL acetone has a volume < 200 mL
B) Roult's law is not obeyed by this system
C) Heat must be absorbed in order to produce the solution at 35°C
D) CS_2 and acetone are less attracted to each other than to themselves

Solution : (Correct Answer: A)

The vapour pressure of mixture (= 600 mm Hg) is greater than the individual vapour pressure of its constituents (Vapour pressure of $CS_2 = 512$ mm Hg, acetone = 344 mm Hg).

Hence, the solution formed shows positive deviation from Raoult's law.

(1) $\Delta_{sol.}H > 0$,

(2) Raoult's law is not obeyed

(3) $\Delta_{sol.}$ Volume > 0

(4) CS_2 and Acetone are less attracted to each other than to themselves.

81. The vapour pressure of pure liquid A decreases from 0.80 atm to 0.6 atm on mixing a non-volatile solute B to liquide A . The mole fraction of B in the solution is

A) 0.25 B) 1 C) 0.50 D) 0.75

Solution : (Correct Answer: A)

According to Raoult's Law

$\frac{P^0 - P_s}{P^0} = x_B$ (Mole fraction of solute)

$x_B = \frac{0.8-0.6}{0.8} = 0.25.$

82. Urea $(NH_2 - CO - NH_2)$ needs to be dissolved in 100g of water, in order to decrease the vapour pressure of water by 25%. What will be the molality of the solution?

A) 18.52 m B) 62.45 m C) 28.52 m D) 35.64 m

Solution : (Correct Answer: A)

$\frac{p_0 - p_S}{p^0} = x_{Solute}$

$= \frac{\frac{W_{Solute}}{M_{Solute}}}{\frac{W_{Solute}}{M_{Solute}} + \frac{W_{Solvent}}{M_{Solvent}}}$

$\frac{P^0 - P_S}{P_0} = \frac{w_A}{m_A} \times \frac{m_B}{W_B}$ (A stands for solute and B for solvent)

$\frac{x - x \times \frac{75}{100}}{x \times \frac{75}{100}} = \frac{w}{60} \times \frac{18}{100}$

[Let the vapour pressure of pure water, $P^0 = x$]

$\frac{0.25x}{0.75x} = \frac{w}{60} \times \frac{18}{100}$

$\therefore\ w = 111.118$

$\text{Molality} = \frac{111.11}{60} \times \frac{1000}{100} = 18.52$

83. In which of the following solution the depression in freezing point is lowest?

A) 0.2 M urea and 0.2 M glucose
B) 0.1 M $Al_2(SO_4)_3$ and 0.1 M Na_2SO_4
C) 0.1 $MKNO_3$ and 0.2 M $Ba(NO_3)_2$
D) 0.1 M $Ca(NO_3)_2$ and 0.1 M $Ba(NO_3)_2$

Solution : (Correct Answer: A)

As $\Delta T_f \propto m \times i$

Urea and glucose do not dissociate in aqueous medium, hence their solution has lowest depression in freezing point as for them i is only 1. It will be maximum for 0.1 M $Al_2(SO_4)_3$ and 0.1 M Na_2SO_4 as $m \times i$ will be maximum for them.

$\Delta T_f \propto i \times m$

$T_f \propto \frac{1}{i \cdot m}$

$\therefore\ \Delta T_f = 0 = T_f$

84. Dissolution of 1.5 g of non-volatile solute (Molecular weight = 60) in 250 g of a solvent reduces its freezing point by 0.01^oC . Find the molal depression constant of the solvent.

A) 0.01 B) 0.001 C) 0.0001 D) 0.1

Solution : (Correct Answer: D)

Depression in freezing point,

$\Delta T_f = k_f \times m$

Where,

$m = \text{Molality} = \frac{\text{Weight of solute} \times 1000}{\text{Molecualr weight of solute} \times \text{Weight of solvent}}$

$= \frac{1.5 \times 1000}{60 \times 250}$

$\Delta T_f = 0.1^\circ \text{ C molal}^{-1}$

$\Rightarrow \Delta T_f = k_f \times 0.1$

$0.01 = k_f \times 0.1$

$\therefore \ k_f = \frac{0.01}{0.1}$

$= 0.1$

85. Insulin $(C_6H_{10}O_5)_n$ dissolved in a medium shows osmotic pressure π at C gm/ cm^3 concentration and 300 K temperature. The slope of plot of π against C is 4.1×10^{-3} . Molecular mass of insulin is

A) 3×10^3 B) 6×10^6 C) 3×10^6 D) 6×10^3

Solution : (Correct Answer: B)

$\frac{\pi V}{1000} = nRT \quad (V \text{ in } Cm^3)$

$\pi V = \left(\frac{w_B \times 1000}{M_B}\right) RT$

$\pi = \left(\frac{w_B}{V}\right) \frac{1000RT}{M} = C \times \frac{1000RT}{M_B}$

When π is plotted against C, the slope will be equal to $\frac{1000RT}{M_B}$.

So $4.1 \times 10^{-3} = \frac{1000 \times 0.082 \times 300}{M_B}$

$M_B = 6 \times 10^6$.

86. Which one of the following statements regarding Henry's law is not correct?

A) The value of K_H changes with function of the nature of the gas.

B) Higher the value of K_H at a given pressure, higher is the solubility of the gas in the liquids

C) The partial pressure of the gas in vapour phase is proportional to the mole fraction of of the gas in the solution.

D) Different gases have different K_H (Henry's law constant) value at a same temperature.

Solution : (Correct Answer: B)

Liquid solution $P_{gas} = K_H \times X_{gas}$ as $K_H \propto \frac{1}{X_{gas}}$

So, more is K_H less is solubility. Solubility of gases is less at higher temperature. So more is the temperature more is the K_H as $K_H \propto \frac{1}{\text{solubility}} \propto$ temperature

87. For a dilute solution containing 2.5 g of a non-volatile non–electrolyte solute in 100 g of water, the elevation in boiling point at 1 atm pressure is $2^\circ C$. Assuming concentration of solute is much lower than the concentration of solvent, the vapour pressure (in mm of Hg) of the solution is (take $K_b = 0.76 \text{ Kkg mol}^{-1}$)

A) 718 B) 736 C) 724 D) 740

Solution : (Correct Answer: C)

$\Delta T_f = iK_f m$ ($i = 1$ for non-electrolyte)

$2 = 0.76 \times \frac{n}{\frac{100}{1000}}$

$n = \frac{2}{7.6}$

Relative lowering in vapour pressure $= X_B$ (mole fraction of solute)

$\frac{\Delta P}{P_A^0} = \frac{2/7.6}{100/18}$

$\frac{\Delta P}{760} = \frac{36}{760}$

$\Delta P = 36$

$P = 760 - 36$

$P = 724 \text{ mm of Hg}$

88. The vapour pressure of water at $20^\circ C$ is 17.5 mmHg.

If 18 g of glucose $(C_6H_{12}O_6)$ is added to 178.2 g of water at $20^\circ C$, the vapour pressure of the resulting solution will be

A) 17.675 mmHg B) 15.750 mmHg
C) 16.500 mmHg D) 17.325 mmHg

Solution : (Correct Answer: D)

Moles of glucose $= \frac{18}{180} = 0.1$

Moles of $H_2O = \frac{178.2}{18} \ \ 9.9$

According to Raoult's law

$\frac{P^\circ - P_s}{P^\circ} = X_{solute}$

$\frac{17.5 - P_S}{17.5} = \frac{0.1}{10}$

so, $P_s = 17.325 \text{ mm Hg}$

89. The factors on which degree of ionization of a compound depends is

A) Size of solute molecules

B) Quantity of electricity passed

C) Nature of vessel used

D) Nature of solute molecules

Solution : (Correct Answer: D)

The degree of ionisation of a solute depends upon its nature, concentration, and temperature.

90. pH of 0.1 M BOH (weak base) is found to be 12. The solution at temperature T K will display an osmotic pressure equal to:

A) 0.01 RT B) $0.01(\text{RT})^2$ C) 0.11 RT D) 1.1 RT

Solution : (Correct Answer: C)

$pH = 12, \ pOH = 14 - 12 = 2,$

$[OH^-] = 10^{-2} = 0.1\ \alpha$

or $\alpha = 0.1$

$i = 1 + (n-1)\alpha = 1 + (2-1) \times 0.1 = 1.1$

$\pi = iCRT = 1.1 \times 0.1\ RT = 0.11\ RT$

Biology - (Zoology)

91. Which of the following can be complications resulting from $STDs$ without timely detection and treatment?
A. Pelvic inflammatory disease (PID)
B. Still births
C. Infertility or even cancer of reproductive tract
D. Ectopic pregnancies

A) A & $B only$
B) A, B & C
C) $A\ only$
D) A, B, C & D

Solution : (Correct Answer: D)
Still births refer to birth of dead baby.
Ectopic pregnancies refer to implantation of blastocyst in parts of female reproductive system other than uterus. Uterus may be infected in various $STDs$

92. Match List I with List II

List I	List II
A. Non-medicated IUD	I. Multiload 375
B. Copper releasing IUD	II. Progestogens
C. Hormone releasing IUD	III. Lippes loop
D. Implants	IV. $LNG-20$

Choose the correct answer from the option given below:

A) $A-I, B-III, C-IV, D-II$
B) $A-IV, B-I, C-II, D-III$
C) $A-III, B-I, C-IV, D-II$
D) $A-III, B-I, C-II, D-IV$

Solution : (Correct Answer: C)
Correct answer is option (3) because

- Lippes loop is a non-medicated IUD.

- Multiload 375 is a copper releasing IUD.

- $LNG-20$ is a hormone releasing IUD.

- Progestogens are used as implants.

93. Which of the following $STDs$ cannot be treated with antibiotics?
A. Gonorrhoea B. Syphilis C. Chlamydia D. Genital herpes

A) $D\ only$
B) $B\ and\ D$
C) $C\ and\ D$
D) $C\ only$

Solution : (Correct Answer: A)
Genital herpes is a viral disease (STD).

94. Match the following columns

Column$-I$	Column$-II$
(A) Chlamydiasis	(1) Chlamydia trachomatis
(B) Gonorrhoea	(2) Neisseria gonorrhoeae
(C) Trichomoniasis	(3) Trichomonas vaginalis
(D) Genital herpes	(4) Herpes simplex virus
(E) Syphilis	(5) Treponema pallidum

A) $(A)-(2).\ (B)-(5),\ (C)-(3),\ (D)-(1),\ (E),\ (4)$
B) $(A)-(4).\ (B)-(5),\ (C)-(3),\ (D)-(2),\ (E),\ (5)$
C) $(A)-(1).\ (B)-(2),\ (C)-(3),\ (D)-(4),\ (E),\ (5)$
D) $(A)-(1).\ (B)-(2),\ (C)-(4),\ (D)-(3),\ (E),\ (5)$

Solution : (Correct Answer: C)
STD
Chalmydiasis - Chlamydia trachomatis
Gonorrhea - Neisseria gonorrhoeae
Trichomoniasis - Trichomonas vaginalis
Genital herpes - Herpes simplex virus
Syphilis - Treponema pallidum

95. Misuse of amniocentesis results in

A) Blood Mismatch
B) Death Of Foetus
C) Genetic Disorders
D) None of these

Solution : (Correct Answer: B)
Misuse of Amniocentesis It is being used to kill the normal female foetus. It is legally banned for the determination of sex to avoid female foeticide

96. Our laws permit legal .... and it is as yet, one of the best methods for couples looking for parenthood.

A) IUI
B) $GIFT$
C) $ZIFT$
D) Adoption

Solution : (Correct Answer: D)

97. Among the following methods, which one has the highest failure rate?

A) Diaphragm with spermicide
B) Condom
C) Intrauterine device
D) Rhythm method

Solution : (Correct Answer: D)
Rhythm method is periodic abstinence during fertile period (from day 10 to 17 of the menstrual cycle when ovulation could be expected) others are barrier methods

98. Implants under the skin and injectables contain

A) Progestogen alone
B) Progestogen and estrogen
C) $FSH\ and\ LH$
D) Both (a) & (b) are correct

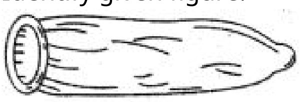
Solution : (Correct Answer: D)
Norplant is progestin only implant.

99. 'family planning' program was initiated in ....

A) 1950 B) 1951 C) 1952 D) 1953

Solution : (Correct Answer: B)

100. Identify given figure.



A) Condom for female B) Loop
C) CuT D) Condom for male

Solution : (Correct Answer: D)

101. What is the full form of $ZIFT$?

A) Zygote Induce Fallopian Transfer
B) Zygote Intra Fallopian Transfer
C) Zygote Inert Fallopian Transfer
D) Zygote In Fallopian Transfer

Solution : (Correct Answer: B)

102. Which oral contraceptive is developed by $CDRI$?

A) Saheli B) Mala$-D$
C) Both (a) and (b) D) None of these

Solution : (Correct Answer: A)

Saheli. India Research in Reproductive Health It should be encouraged and supported to find out the new methods in reproduction related areas. 'Saheli' a new oral contraceptive for the females was developed by scientists in Central Drug Research Institute (CDRI) in Lucknow

103. What is the purpose of surgical method of contraception?

A) Prevent gamete motility
B) Prevent gamete formation
C) Gametogenesis promotion
D) Facilitate implantation

Solution : (Correct Answer: A)

Surgical Method of Contraception In that method the cutting of vas deferens in male and Fallopian tubes in female takes place due to which the motility of gametes (ova and sperm) inhibited. Vasectomy male semen have all the constitutes (secretion of Cowber's glands, seminal vesicle and prostate gland) but don't have gametes (sperm)

104. Ovulation do not occur in lactational period because of

A) Inhibin B) Prolactin
C) Prostaglandin D) Oxytocin

Solution : (Correct Answer: B)

In lactating mother, there is the release and the production of milk secreting hormone. These hormones suppresses the release of Follicle Stimulating Hormone (FSH), so during intense lactation there is no ovulation hence, no pregnancy

105. Identify the correct statements

i. Birth control pills are likely to cause cardiovascular problem
ii. A woman who substitutes or takes the place of the real mother to nurse to embryo is called surrogate mother
iii. Numerous children have been produced by in vitro fertilisation but with some abnormalities
iv. Woman plays a key role in the continuity of the family and human species
v. Foetal sex determination test should not be banned

A) I and II
B) II and IV
C) III and V
D) I, II and IV

Solution : (Correct Answer: D)

Oral contraceptive pills increases the risk of intra vascular clotting. Therefore, they are not recommended for women with a history of disorders of blood clotting, careful blood vessel damage, hypertension, liver malfunction, heart disease or cancer of the breast or reproductive system

106. Which of the following is an incorrect statement for periodic abstinence?

A) The couple should abstain from coitus from day 10 to 17 of the menstrual cycle when ovulation could be expected
B) 10^{th} to 17^{th} day of the cycle is fertile period, when the chances of fertilisation are high
C) This prevents the chances of union of male and female gametes
D) In this method, the ovum and sperms are prevented from physically meeting with the help of barriers

Solution : (Correct Answer: D)

Periodic abstinence is a natural method, not a barrier method.

107. Which of the following is world’s first non-hormonal oral contraceptive pill for females, developed by scientists at Central Drug Research Institute $(CDRI)$ in Lucknow, India?

A) Mala$-D$ B) Saheli
C) Morning after pills D) PoP

Solution : (Correct Answer: B)

Saheli is a non steroidal drug that works by preventing implantation.

108. Birth rate is directly proportional to

A) Good food B) Illiteracy
C) High social status D) Age of marriage

Solution : (Correct Answer: B)

It's Obvious

109. One of the legal methods of birth control is

A) by having coitus at the time of day break
B) by a premature ejaculation during coitus
C) by abstaining from coitus from day 10 to 17 of the menstrual cycle.

D) abortion by taking an appropriate medicine

Solution : (Correct Answer: C)

(c) :

By abstaining from coitus from 10-17 days of menstrual cycle , during which the egg is released , the pregnancy can be prevented. This method of birth control is called 'periodic abstinence'. This is one of the effective method to prevent pregnancy and is legal.So, the correct answer is 'By abstaining from coitus fom 10-17 days of menstrual cycle'.

110. At present total population of the world is about

A) 4.5 billions B) 5.2 billions
C) 6.2 billions D) None of the above

Solution : (Correct Answer: C)

(c) The population of the world has steadily increased to 5 billion on July $11, 1987$ and 6 billion on October $12, 1999$ and 6.2 billion on 2001.

111. Trade name of weekly oral contraceptive pill is

A) Mala B) Saheli C) Mala A D) Mala D

Solution : (Correct Answer: B)

It's Obvious

112. Tubectomy is a method of sterilization in which

A) small part of the Fallopian tube is removed or tied up
B) ovaries are removed surgically
C) small part of vas deferens is removed or tied up
D) uterus is removed surgically.

Solution : (Correct Answer: A)

(a) : Sterilization provides a permanent and sure birth control. In females, it is called tubectomy. Tubectomy involves the blocking of the Fallopian tubes. A small part of the Fallopian tube is removed or tied up through a small incision in the abdomen or through vagina.

113. Which gland's secretion is act as lubricants in male?

A) Bulbourethral gland B) Pancrease
C) Seminal vesicle D) Vas deferens

Solution : (Correct Answer: A)

114. Which organ of human is not paired ?

A) Vas deferens B) Prostate gland
C) Bulbourethral gland D) Testis .

Solution : (Correct Answer: B)

115. Which of the following hormones is not a secretory product of human placenta?

A) Human chorionic gonadotropin B) Prolactin
C) Oestrogen D) Progesterone

Solution : (Correct Answer: B)

Prolactin is secreted by anterior pituitary gland, which stimulates mammary gland development during pregnancy and lactation after child birth.

116. No new follicles develop in the luteal phase of the menstrual cycle because

A) Follicles do not remain in the ovary after ovulation
B) FSH levels are high in the luteal phase
C) LH levels are high in the luteal phase
D) Both FSH and LH levels are low in the luteal phase

Solution : (Correct Answer: D)

117. Ejaculation of human male contains about $200 - 300$ million sperms, of which for normal fertility ____ % sperms must have normal shape and size and at least ____% must show energetic motility.

A) $40, 60$ B) $50, 50$
C) $60, 40$ D) $30, 70$

Solution : (Correct Answer: C)

Ejaculation of human male contains about $200 - 300$ million sperms, of which for normal fertility 60% sperms must have normal shape and size and at least 40% must show energetic motility.

118. The correct sequence of spermatogenetic stages leading to the formation of sperms in a mature human testis is

A) spermatogonia - spermatocyte - spermatid - sperms
B) spermatid - spermatocyte - spermatogonia - sperms
C) spermatogonia - spermatid - spermatocyte - sperms
D) spermatocyte - spermatogonia - spermatid - sperms.

Solution : (Correct Answer: A)

(a) : Spermatogenesis is the process of formation of haploid spermatozoa (sperms) from diploid spermatogonia inside the testes of the male. At sexual maturity, the undifferentiated primordial germ cells divide several times by mitosis to produce a large number of spermatogonia or sperm mother cells. Each spermatogonium actively grows to a larger primary spermatocyte by obtaining nourishment from the nursing cells. The phenomenon of formation of primary spermatocytes from spermatogonia, is called spermatocytogenesis. Each primary spermatocyte undergoes two successive divisions, called maturation divisions. The first maturation division is reductional or meiotic. Hence, the primary spermatocyte divides into two haploid daughter cells called secondary spermatocytes. Both secondary spermatocytes now undergo second maturation division which is an ordinary mitotic division to form four haploid spermatids, by each primary spermatocyte. The transformation of spermatids into spermatozoa is called spermiogenesis or spermateleosis or differentiation phase.

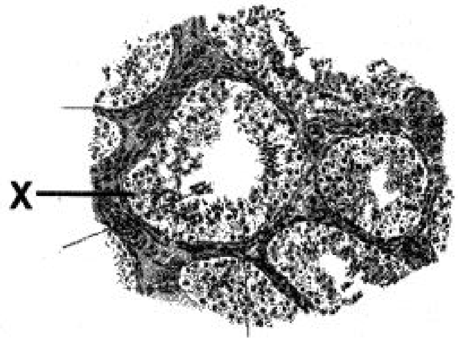
119. In human adult females oxytocin

A) stimulates pituitary to secrete vasopressin
B) causes strong uterine contractions during parturition
C) is secreted by anterior pituitary
D) stimulates growth of mammary glands.

Solution : (Correct Answer: B)

(b) : In human adult female oxytocin is a hormone released by the pituitary gland (neurohypophysis), that causes contraction of the uterus during labour and stimulates milk flow from the breasts by causing contraction of muscle fibres in the milk ducts.

120. Identify X from figure



A) Interstitial cells
B) Spermatogonia
C) Langerhans cell mass
D) Sertoli cells

Solution : (Correct Answer: B)

121. If the vas deferens of a man is surgically disconnected

A) Sperms in the semen will be without nuclei
B) Semen will be without sperms
C) Spermatogenesis will not occur
D) Sperms in the semen will be non-motile

Solution : (Correct Answer: B)

(b) Function of vasa deferentia are conduction of sperms by peristalsis of its highly muscular coat.

It disconnected in man so semen will be without sperms.

122. The placenta helps in the maintenance of pregnancy by secreting a hormone, known as

A) Thyroxine
B) Estrogen
C) Progesteron
D) Testosterone

Solution : (Correct Answer: C)

(c)Progesterone serves to maintain pregnancy by preventing contraction of uterine wall.

123. If after ovulation no pregnancy results, the corpus luteum

A) Is maintained by the presence of progesterone
B) Degenerates in a short time
C) Becomes active and secretes lot of FSH and LH
D) Produces lot of oxytocin and relaxin

Solution : (Correct Answer: B)

It's Obvious

124. The process by which ova are formed is knwon as

A) Oogenesis
B) Ovulation
C) Oviposition
D) Oviparity

Solution : (Correct Answer: A)

(a)The process of formation of haploid ova from diploid germinal cells $(2n)$ of the ovary called oogenesis.

125. Fertilization is depicted by the condition

A) $n \rightarrow 2n$
B) $2n \rightarrow 3n$
C) $2n \rightarrow 4n$
D) $4n \rightarrow 8n$

Solution : (Correct Answer: A)

(a)On fertilization egg becomes diploid $(2n)$.

126. Which one of the following statements with regard to embryonic development in humans is correct

A) Cleavage division bring about considerable increase in the mass of protoplasm
B) In the second cleavage division, one of the two blastomeres usually divides a little sooner than the second
C) With more cleavage divisions, the resultant blastomeres become larger and larger
D) Cleavage division results in a hollow ball of cells called morula

Solution : (Correct Answer: B)

It's Obvious

127. Gastrulation is the process which involves the differentiation of the following layers in a vertebrate embryo

A) Ectoderm and mesoderm
B) Ectoderm and endoderm
C) Endoderm and mesoderm
D) Ectoderm, endoderm and mesoderm

Solution : (Correct Answer: D)

(d)Gastrulation is the process of the embryonic development during which cell movements establish the three primary germinal layers namely ectoderm, mesoderm and endoderm.

128. Ovulation occurs

A) Alternately from two ovaries
B) Simultaneously from both the ovaries
C) From one ovary alone throughout the life
D) According to the season from two ovaries

Solution : (Correct Answer: A)

It's Obvious

129. The male sex hormone 'testosterone' is secreted by special Leydig's cells present in

A) Kidneys

B) Seminiferous tubules of testes
C) Connective tissue surrounding the seminiferous tubules of testes
D) Cells lining the epidermis of the testes

Solution : (Correct Answer: C)

(c) Between seminiferous tubules there are groups of polyhedral endocrine cells called interstitial or leydig's cells.

These secrete a steroid male sex hormone testosterone.

130. Functions of seminal fluid is/are

A) Maintains the viability of sperms
B) Maintains motility of sperms
C) Provides proper pH and ionic strength
D) All the above

Solution : (Correct Answer: D)

It's Obvious

131. The endometrium is the lining of

A) Bladder B) Vagina C) Uterus D) Oviduct

Solution : (Correct Answer: C)

(c) Lining layer of uterus called endometrium $(mucousmembrane)$ is richly supplied with blood vesseles and tubular gland.

Actual wall of uterus is myometrium.

It is covered on outside by perimetrium.

132. Cervix lies between

A) Oviduct and uterus B) Uterus and vagina
C) Vagina and clitoris D) Clitoris and labia

Solution : (Correct Answer: B)

(b) Cervix is lower narrow part which opens in body of uterus by internal os and in vegina below by external os.

133. The cavity present in the graafian follicle is

A) Amniotic cavity B) Archenteron
C) Antrum D) Ostium

Solution : (Correct Answer: C)

(c) The cavity of graffian follicle is antrum or follicular cavity having liquor folliculi and an eccentrically placed oocyte.

134. Fertilization occurs in human, rabbit and other placental mammals in

A) Ovary B) Uterus C) Fallopian tubes D) Vagina

Solution : (Correct Answer: C)

(c) In mammals $(Rabbitandhumanbeings)$, fertilization of the ovum occurs in fallopian tube or oviduct or uterine tube.

135. During the course of development, cells in various regions of embryo become variable in morphology and eventually perform diverse functions. This process is known as

A) Rearrangement B) Differentiation
C) Metamorphosis D) Organisation

Solution : (Correct Answer: B)

It's Obvious

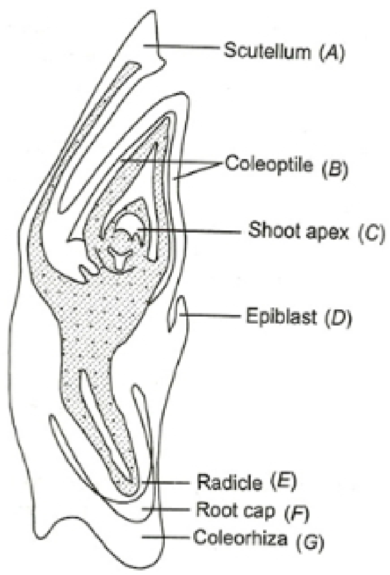
Biology - (Botany)

136. Scutellum is

A) Cotyledon in dicots
B) Cotyledon in gymnosperm
C) Monocot root
D) Cotyledon in grass family

Solution : (Correct Answer: D)

Embryos of monocotyledons possess only one cotyledon. In the grass family the cotyledon is called scutellum that is situated toward the one side (lateral) of the embryonal axis. At its lower end, the embryonal axis has the radical and root cap enclosed in an undifferentiated sheath called coleorhiza.The portion of the embryonal axis above the level of attachment of scutellum is epicotyl. Epicotyl has a shoot apex and few leaf primordial enclosed in hallow structure the coleoptile



LS of an embryo of grass

137. Parthenogenesis is a type of

A) Sexual reproduction
B) Asexual reproduction
C) Budding
D) Regeneration

Solution : (Correct Answer: B)

Parthenogenesis is a type of asexual reproduction because it involves an unfertilized egg cell only:

138. Function of micropyle is

A) Helps in germination
B) Helps in surviving
C) Both (a) and (b)
D) Helps in endosperm formation

Solution : (Correct Answer: C)

Micropyle is the small aperture through, which the water goes inside at the time of germination. It also helps in the gaseous exchange

139. The innermost wall layer of anther

A) Is nutritive in function
B) Helps in dehiscence of anther
C) Is haploid and protective in function
D) Forms microspores

Solution : (Correct Answer: A)
Tapetum is nutritive.

140. In $40\ \%$ angiosperms, the pollen grains are shed at

A) Four-celled stage B) Three-celled stage
C) Two-celled stage D) Five-celled stage

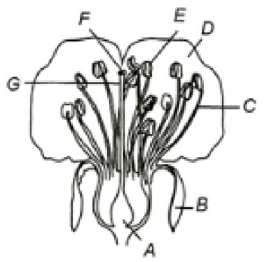
Solution : (Correct Answer: B)
At 3 cell stage pollen grain is liberated in $40\ \%$ of angiosperms
$3-$ Celled stage is a mature gametophy

141. The pollen viability period of rice and pea respectively, is

A) $30\ minutes$ and several months
B) Several months and $30\ minutes$
C) Few days and few months
D) Few days in both the cases

Solution : (Correct Answer: A)

142. Identify A to G in following figure and answer accordingly



A) $A-$Ovary, $B-$Filament, $C-$Sepal, $D-$Petal, $E-$Style, $F-$Stigma, $G-$Anther
B) $A-$Petal, $B-$Ovary, $C-$Petal, $D-$Filament, $E-$Anther, $F-$Stigma, $G-$Style
C) $A-$Ovary, $B-$ Sepal, $C-$ Filament, $D-$ Petal, E $-$Anther, $F-$Stigma, $G-$Style
D) $A-$Ovary, $B-$Sepal, $C-$ Filament, $D-$ Petal, E $-$Anther, $F-$Stigma, $G-$Style

Solution : (Correct Answer: C)
A

143. Sexual reproduction leads to

A) Genetic recombination B) Polyploidy
C) Aneuploidy D) euploidy

Solution : (Correct Answer: A)
Sexual reproduction includes syngamy and meiosis. Syngamy is the nuclear fusion of male and female gamete. The meiosis is reduction of chromosome number to haploid during meiosis. Genetic recombination occurred as a result of crossing over.

144. Milky water of tender coconut is

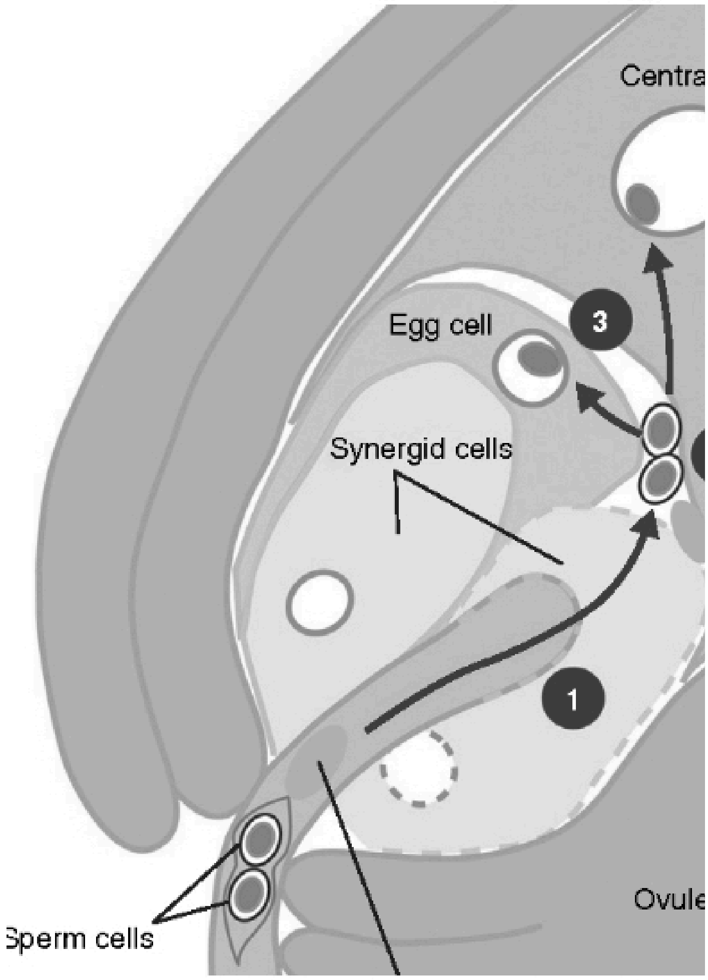
A) Liquid gametes
B) Liquid nucellus
C) Liquid female gametophyte
D) Liquid endosperm

Solution : (Correct Answer: D)
Milky water of tender coconut is called liquid endosperm.

145. The number of female nuclei involved in double fertilization is

A) 2 B) 3 C) 4 D) 1

Solution : (Correct Answer: B)
Two polar nuclei one egg cell $(total\ 3\ nuclei)$

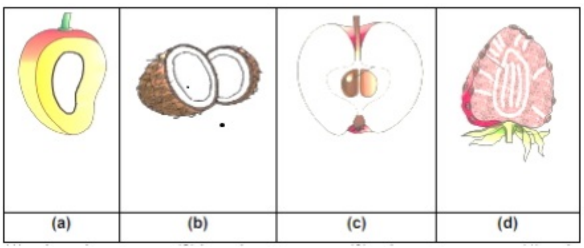


146. Flower is a

A) Modified male plant only
B) Modified female plant only
C) Modified reproductive shoot
D) Vegetative shoot system

Solution : (Correct Answer: D)
Flower is a modified shoot meant for reproduction

147. Identify which of the following fruits are false fruit?



A) a, b, c, d B) b, c, d

C) a, b, c D) c, d

Solution : (Correct Answer: D)

The false fruit have some of their flesh derived from ovary and some of them from other tissue. Apples and strawberries are known as false fruits.

148. Which is incorrect statement?

I. Each cell of sporogenous tissue in anther is capable of giving rise to microspore tetrad.
II. The pollen grain represent male gametophyte.
III. Pollen grains are usually triangular and $10 - 15\mu m$ in diameter.
IV. Sporopollenin is one of the most resistance organic material which can be destroyed only by strong acids
and alkali.

A) I, II are incorrect but III, IV are correct
B) III, IV are incorrect but I, II are correct
C) I, III are incorrect but II, IV are correct
D) II, IV are correct but I, III are incorrect

Solution : (Correct Answer: B)

Pollens are spherical and average size is $25 - 50\mu m$. Sporopollenin donot get destroyed by any known chemical.

149. Mark the incorrect statement

A) Outer three layers of anther wall are protective in function
B) Sporogenous tissue, occupies the centre of each microsporangium
C) Cells of tapetum and endothecium show increase in DNA contents by endomitosis and polyteny
D) Ploidy level of microspore tetrad is haploid

Solution : (Correct Answer: C)

Cells of endothecium will not show increase in DNA content.

150. Which of the following plant provides safe place to insect for laying eggs?

A) Sage plant B) Amorphophallus
C) Ophrys D) Mango

Solution : (Correct Answer: B)

Tallest flowers $\rightarrow$ Which give space for egg laying.

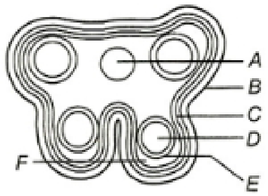
151. What does the filiform apparatus do at the entrance into ovule?

A) It helps in the entry of pollen tube into a synergid
B) It prevents entry of more than one pollen tube into the embryo sac
C) It brings about opening of the pollen tube
D) It guides pollen tube from a synergid to egg

Solution : (Correct Answer: D)

Synergid cells are characterized by the presence of finger like projections called filiform apparatus attached to their upper wall at micropyler end. This filiform apparatus is known to attract and guide the pollen tube.

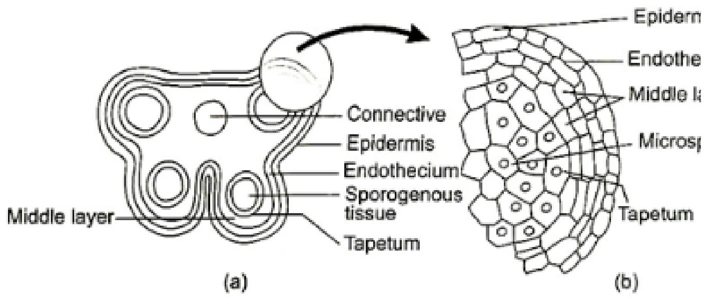
152. Identify A to E in the following diagram



A) $A-$Epidermis, $B-$Endodermis, $C-$Connective tissues, $D-$Sporogenous tissue, $E-$Middle layer, $F-$Tapetum
B) $A-$Endodermis, $B-$ Connective tissues, $C-$ Epidermis, $D-$ Tapetum, $E-$Sporogenous tissue, $F-$ Middle layer
C) $A-$ Tapetum, $B-$ Middle layer, $C-$ Sporogenous tissue, $D-$ Connective tissues, E $-$Endodermis, $F-$Epidermis
D) $A-$ Connective tissues, $B-$ Epidermis, $C-$ Endothecium, $D-$Sporogenous tissue, $E-$ Tapetum, $F-$ Middle layer

Solution : (Correct Answer: D)

$A-$ Connective tissues, $B-$ Epidermis, $C-$ Endothecium, $D-$Sporogenous tissue, $E-$ Tapetum, $F-$ Middle layer Microsporangium is mainly surrounded by four layers/wall, i.e., Epidermis, endothecium, middle layer and tapetuem (i) Epidermis endothecium and middle layer help in protection and dehiscence of anther from pollen (ii) Tapetum nourishes the developing pollen grains



(a) Transverse section of a young other
(b) Enlarged view one micro sporangium showing

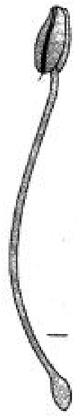
153. Which one of the following statements is correct?

A) Endothecium produces the microspores.
B) Tapetum nourishes the developing pollen.
C) Hard outer layer of pollen is called intine.
D) Sporogenous tissue is haploid.

Solution : (Correct Answer: B)

(b) : A microsporangium is generally surrounded by four wall layers -the epidermis, endothecium, middle layers and the tapetum. The outer three wall layers perfom the function of protection and help in dehiscence of anther to release the pollen. The innermost wall layer is the tapetum. It nourishes the developing pollen grains. Cells of the tapetum are food rich and possess dense cytoplasm and generally have more than one nucleus. They disintegrate to liberate the contents which is absorbed by the developing spores.

154. Identify figure.



A) Typical gynoecium B) Typical stamen
C) Cut section of microsporangium D) Flower

Solution : (Correct Answer: B)

155. Which one of the following is resistant to enzyme action?

A) Pollen exine B) Leaf cuticle
C) Cork D) Wood fibre

Solution : (Correct Answer: A)

(a) : Sporopollenin is a major component of the tough outer (exine) walls of spores and pollen grains. It is chemically very stable and is usually well preserved in soils and sediments. It can withstand environmental extremes and cannot be degraded by enzymes and strong chemical reagents.

156. Nucellar polyembryony is reported in species of

A) Citrus B) Gossypium C) Triticum D) Brassica.

Solution : (Correct Answer: A)

(a) : In nucellar polyembryony, some of the nucellar cells surrounding the embryo sac start dividing. Then it protrudes into the embryo sac and develop into the embryos. In such species, each ovule contains many embryos. Occurrence of more than one embryo in a seed is referred as polyembryony. Nucellar polyembryony is found in many of the Citrus and mango varieties.

157. Both, autogamy and geitonogamy are prevented in

A) papaya B) cucumber C) castor D) maize.

Solution : (Correct Answer: A)

(a) : Autogamy and geitonogamy are two forms of self pollination. In autogamy, pollen falls on stigma of the same flower. While in geitonogamy pollens from a flower fall on the stigma of some other flower on the same plant. Papaya is a dioecious plants thus both autogamy and geitonogamy are prevented in it.

158. Which one of the following statements is correct?

A) Cleistogamous flowers are always autogamous.
B) Xenogamy occursonly bywind pollination.
C) Chasmogamous flowers do not open at all.
D) Geitonogamy involves the pollen and stigma of flowers of different plants.

Solution : (Correct Answer: A)

(a) : In cleistogamy, as the flowers never open so there is no alternative of self pollination. It is invariably autogamous. In xenogamy, pollination takes between two flowers of different plants (genetically and ecologically). It can ocur by wind, water, insects and animals.Chasmogamy occurs when the flowers expose their mature anther and stigma to the pollinating agents. Geitonogamy is the pollination taking place between the two flowers of the same plant or genetically similar plant. Genetically, it is self pollination but as the agency is involved it is ecologically cross pollination.

159. Where does pollen tube developes ?

A) In style
B) In anthor
C) On the stigma
D) In stalk (Filment of anthor)

Solution : (Correct Answer: A)

160. In which of the plants the seeds are thousands?

A) Phoenix dectylyfera B) Mango
C) Orchids D) Carrot

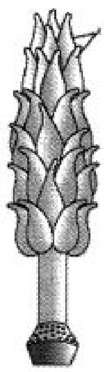
Solution : (Correct Answer: C)

161. It is not included in post fertilisation event

A) Seed formation B) Fruit formation
C) Gametogenesis D) Embryogenesis

Solution : (Correct Answer: C)

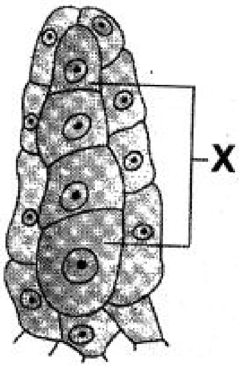
162. Identify the given figure :



A) Flower of michelia
B) Gynoecium of michelia
C) Flower of angiosperm plant
D) Leaf of gymnosperm

Solution : (Correct Answer: B)

163. In given figure, labeled part X is formed by....



A) PMC B) Megaspore Tetrad
C) Microspore D) Megaspore

Solution : (Correct Answer: B)

164. ....pollination is quite common in grasses

A) Water B) Wind C) Insect D) Animal

Solution : (Correct Answer: B)

165. Pollination in water hyacinth and water lily is brought about by the agency of

A) water B) insects or wind C) birds D) bats.

Solution : (Correct Answer: B)

(b) : In aquatic plants with emergent flowers $e.g.$, water lily, water hyacinth pollination takes place by wind or insects.

166. The endosperm of Brassicais

A) Haploid B) Diploid C) Triploid D) Tetraploid

Solution : (Correct Answer: C)

(c)Brassica is a angiosperm plant.

167. Fertilization of egg takes place inside

A) Anther B) Stigma C) Pollen tube D) Embryo sac

Solution : (Correct Answer: D)

(d)Because egg is the part of embryo sac.

168. Cheiropterophily is the process of pollination by

A) Water B) Bat C) Insect D) Bird

Solution : (Correct Answer: B)

It's obvious.

169. Which of the following is without exception in angiosperms

A) Secondary growth
B) Presence of vessels
C) Double fertilization
D) Autotrophic nutrition

Solution : (Correct Answer: C)

(c)Double fertilization is found only in angiosperms. In which secondary nucleus form to triploid cell and egg convert into diploid zygote. Triploid cell to form endosperm and diploid zygote to form embryo.

170. In a seed of maize, scutellum is considered as cotyledon because it

A) Protects the embryo
B) Contains food for the embryo
C) Absorbs food materials and supplies them to the embryo
D) Converts itself into a monocot leaf

Solution : (Correct Answer: C)

It's obvious.

171. The sequence of development of embryo sac is

A) Archesporium $\rightarrow$ megaspore mother cell $\rightarrow$ megaspore $\rightarrow$ embryo sac
B) Archesporium $\rightarrow$ megaspore $\rightarrow$ megaspore mother cell $\rightarrow$ embryo sac
C) Archesporium $\rightarrow$ megaspore $\rightarrow$ megasporophyte $\rightarrow$ embryo sac
D) None of the above

Solution : (Correct Answer: A)

It's obvious.

172. The pollen tube usually enters the embryo sac

A) Between one synergid and central cell
B) By directly penetrating the egg
C) Through one of the synergids
D) By knocking off the antipodal cells

Solution : (Correct Answer: C)

It's obvious.

173. Which of the following pairs of plant parts are both haploid

A) Antipodal cells and egg cells
B) Nucellus and primary endosperm nucleus
C) Nucellus and antipodal cells
D) Antipodal cells and megaspore mother cells

Solution : (Correct Answer: A)

(a)Antipodal cells and egg cells are formed by megaspore.

174. The stalk of the ovule is called

A) Pedicel B) Petiole C) Funicle D) Hilum

Solution : (Correct Answer: C)

It's obvious.

175. In which type whole of the megaspore mother cell takes part in the formation of the female gametophyte

A) Monosporic 8 nucleate
B) Monosporic 4 nucleate
C) Bisporic
D) Tetrasporic

Solution : (Correct Answer: D)

(d)The diploid megaspore mother cell enlarges in size and divides by meiosis to form a linear tetrad of four haploid megaspores. Megaspore is the first cell of female gametophyte.

176. The mature stigma is either rough or sticky in

A) All types of flowers
B) Water pollinated flowers
C) Wind pollinated flowers
D) Insect pollinated flowers

Solution : (Correct Answer: C)

It's obvious.

177. Correct definition of pollination is

A) Transfer of pollen grain from anther to stigma
B) Germination of pollen grain
C) Growth of pollen tube in ovule
D) Visits of insects in flower

Solution : (Correct Answer: A)

It's obvious.

178. In which characters air pollinated flowers differ from insect pollinated ones

A) Due to small parianth and sticky pollen
B) Small coloured parianth and heavy pollen grains
C) Coloured parianth and large pollen grains
D) Without parianth and light pollen grains

Solution : (Correct Answer: D)

It's obvious.

179. Anther is generally composed of

A) One sporangium
B) Two sporangium
C) Three sporangium
D) Four sporangium

Solution : (Correct Answer: D)

(d)A typical anther consist of four microsporangia (tetra sporangiate) and such anther is called dithecous.

180. During meiosis in pollen mother cell the daughter cells are interconnected by passages. The whole structure is called

A) Symplast
B) Plasmodesmata
C) Syncytium
D) Coenocyte

Solution : (Correct Answer: C)

It's obvious.